# Universal Dynamic Portfolios

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**Abstract**

Universal Portfolio [Cover, 1991] matches the best *constant* rebalanced portfolio in hindsight. We extend this framework to compete with an arbitrary *dynamic* portfolio sequence $\mathbf{u}_1, \dots, \mathbf{u}_T$, a dynamic-regret setting for log-loss where existing methods break down due to potentially unbounded gradients. We establish an $\Omega\big(\max\{\log T, T^{1/2}P_T^{1/2}\}\big)$ lower bound, showing that the linear-loss-type $T^{1/2}P_T^{1/2}$ dependence is unavoidable in general despite log-loss exp-concavity. Here, $P_T = \sum_{t=2}^{T}\|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$ measures the total amount of portfolio changes over $T$ rounds. We then introduce an algorithm that achieves the matching minimax-optimal rate for all $\mathbf{u}_{1:T}$ and a faster $\mathcal{O}\big(\max\{\log T, T^{1/3}P_T^{2/3}\}\big)$ when $\mathbf{u}_{1:T}$ remains in the strict interior of the simplex. Our techniques combine a new Dirichlet Hedge algorithm with fixed share. Finally, under an additional condition that daily stock price relatives are bounded away from zero and infinity, yielding bounded gradients, we obtain a computationally efficient online method with fast rates, mirroring the classical shift from Universal Portfolio to efficient algorithms such as Online Newton Step.

Online portfolio selection (OPS), studying how to sequentially allocate wealth among a set of assets to maximize cumulative returns, is a textbook motivating example in online learning [Hazan, 2016]. The study of OPS has significantly influenced online learning with curved loss functions, leading to the development of fundamental algorithmic and analytical tools [Agarwal et al., 2006, Van Erven et al., 2020, Zimmert et al., 2022, Mhammedi and Rakhlin, 2022, Jézéquel et al., 2025]. The problem was first introduced by Cover [1991] as a distribution-free model for sequential investment, who also proposed the seminal Universal Portfolio algorithm attaining an $\mathcal{O}(d \log T)$ regret bound against any constant rebalanced portfolio (CRP) that allocates wealth according to fixed proportions. Subsequently, OPS has attracted substantial attention in the online learning community due to its rich structure, as well as the challenge of designing efficient algorithms beyond Universal Portfolio, whose implementation requires evaluating integrals over the simplex and is therefore inefficient.

A notable progress in OPS is the Online Newton Step (ONS) algorithm proposed by Agarwal et al. [2006], which takes Cover's log-loss as an exp-concave function and achieves an $\mathcal{O}(dG \log T)$ regret bound with $\mathcal{O}(d^3)$ computation per round, where $G$ is an upper bound on the gradient norm and $d$ is the dimension of the portfolio space. However, the dependence on $G$ is undesirable, since Cover's log-loss can be unbounded in the worst case. As posed by Van Erven et al. [2020], an important open question is whether there exists an algorithm for OPS that is simultaneously computationally efficient, achieves fast regret rates, and avoids dependence on the gradient norm. This question has spurred a line of work developing more advanced techniques to address these challenges [Luo et al., 2018, Zimmert et al., 2022, Mhammedi and Rakhlin, 2022, Jézéquel et al., 2025].

However, in a continuously evolving and possibly adversarial market, competing against a constant comparator, including the best CRP, may be restrictive. This motivates us to extend the Universal Portfolio framework for time-varying comparators. This problem naturally fits within the online convex optimization framework with Cover's log-loss, when the *dynamic regret* is taken as the performance measure. Let $\ell_t(\mathbf{w}) = -\ln(\mathbf{w}^\top \mathbf{x}_t)$ be *Cover's loss*, where $\mathbf{w} \in \Delta_d$ denotes the learner's portfolio vector on the probability simplex, and $\mathbf{x}_t \in \mathbb{R}_+^d$ denotes the market return at time $t$. The dynamic regret [Zinkevich, 2003, Zhang et al., 2018] measures the gap between the learner's cumulative loss and

that of a time-varying comparator sequence $\{\mathbf{u}_t\}_{t=1}^T$ by

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) = \sum_{t=1}^T \ell_t(\mathbf{w}_t) - \sum_{t=1}^T \ell_t(\mathbf{u}_t). \tag{1}$$

The above measure is often referred to as universal dynamic regret, as it seeks guarantees that are valid for any comparator sequence, with regret bounds that usually scale with the *path-length* $P_T = \sum_{t=2}^T \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$ measuring the variation of the sequence. Dynamic regret reduces to the classical notion of (static) regret by setting $\mathbf{u}_t = \mathbf{w}_* = \arg\min_{\mathbf{w}\in\Delta_d} \sum_{t=1}^T \ell_t(\mathbf{w})$. At the same time, it offers greater flexibility by allowing the comparator sequence $\mathbf{u}_t$ to adapt to the underlying environment, rather than being tied to a single realized return. For instance, when the market return $\mathbf{x}_t$ is sampled from a time-varying distribution $\mathcal{D}_t$, a natural choice is $\mathbf{u}_t = \arg\min_{\mathbf{w}\in\Delta_d} \mathbb{E}_{\mathbf{x}_t\sim\mathcal{D}_t}[\ell_t(\mathbf{w})]$. Compared with the minimizer $\mathbf{w}_t^\star = \arg\min_{\mathbf{w}\in\Delta_d} \ell_t(\mathbf{w})$ at each round, this choice avoids chasing noise from a single observation.

**Related Work and Challenge.** Although non-stationary online learning has been extensively studied over the past decades [Herbster and Warmuth, 1998, Hazan and Seshadhri, 2009, Cesa-Bianchi et al., 2012, György and Szepesvári, 2016, Zhang et al., 2018, Baby and Wang, 2021, Wei and Luo, 2021, Zhao et al., 2024], it is surprisingly underexplored for OPS. To the best of our knowledge, the only non-stationary OPS result that avoids dependence on $G$ is due to Singer [1997], who proposed a method that competes with comparators switching among $N$ fixed portfolios. This corresponds to a restricted notion of dynamic regret, in which $\mathbf{u}_t$ is confined to a finite set of points and cannot evolve continuously over time, making the problem substantially more tractable.

Even under the bounded-gradient norm assumption, no fast-rate results are known for the non-stationary setting. The most closely related techniques come from non-stationary online exp-concave optimization (OXO) [Baby and Wang, 2021, Zhang et al., 2025], which show that curvature in the loss function, such as exp-concavity or strong convexity, leads to faster learning rates than linear or general convex losses, even in non-stationary environments. However, these results do not directly apply to OPS due to the issue of *domain constraints*, an additional challenge posed for non-stationary OPS. We provide a more detailed discussion about this in Section 1.

Somewhat surprisingly, the best-known computationally tractable algorithm for non-stationary OPS remains that of Zhang et al. [2018], which is designed for general convex losses and achieves a dynamic regret of $\mathcal{O}(G\sqrt{T(1+P_T)})$. This bound incurs a linear dependence on $G$ and, even in the bounded-gradient case, is strictly weaker than those available for exp-concave losses, as it fails to exploit the structure of the log loss. This naturally raises the following question:

*What is the minimax dynamic regret bound for non-stationary online portfolio selection?*

**Our Results.** In this paper, we provide several algorithms with upper bounds and lower bound to characterize the behavior of OPS in non-stationary environments:

- **Lower Bound:** We first establish an $\Omega(\max\{\log T, \sqrt{TP_T}\})$ lower bound for OPS with a two-asset market. This result indicates that curvature in OPS does not substantially improve the $T$-dependence of dynamic regret. In particular, in relatively non-stationary environments, the achievable regret rate is comparable to that of general convex losses in terms of $T$, for which an $\mathcal{O}(G\sqrt{T(1+P_T)})$ bound is known to be optimal. We also provide a simple reduction-based method that achieves a matching $\mathcal{O}(\max\{\log T, \sqrt{TP_T}\})$ upper bound.

- **Achievable Fast Rates:** We then show that a fast-rate dynamic regret is achievable with additional regularity of the comparator $\mathbf{u}_t$. We provide an algorithm that ensures a fast-rate dynamic regret of $\mathcal{O}(\max\{\log T, \alpha^{-1/3}T^{1/3}P_T^{2/3}\})$ when the comparator sequence $\{\mathbf{u}_t\}_{t=1}^T$ lies in the *relative interior* of the simplex with margin $\alpha$, i.e., $\mathbf{u}_t \in \Delta_d^\alpha = \{\mathbf{u} \in \Delta_d : u_i \geq \alpha, \forall i \in [d]\}$. This result follows from a novel mixability-based analysis of Cover's Universal Portfolio algorithm with fixed share updates, which adopts a Dirichlet distribution as the comparator to accommodate the simplex constraint. Moreover, we show that this fast-rate guarantee seamlessly degrades to the minimax-optimal $\mathcal{O}(\max\{\log T, \sqrt{TP_T}\})$ bound that holds uniformly over all $\mathbf{u}_t$, even when $\alpha$ is close to 0.

- **Computationally Efficient Method in the Fast Rate Regime.** When imposing an additional assumption on an upper bound of the gradient norm $G$, we propose an efficient algorithm achieving fast-rate $\mathcal{O}(G \max\{\log T, T^{1/3} P_T^{2/3}\})$ dynamic regret for all comparators. This result applies not only to OPS but also to general OXO over arbitrary convex domains. To our knowledge, this is the *first* computationally tractable method with such generality; prior work [Baby and Wang, 2022a] crucially relies on box constraints for their KKT-based analysis and is hard to extend. Interestingly, our method turns out to be Follow-the-Leading-History [Hazan and Seshadhri, 2009] combined with Online Newton Step [Hazan et al., 2007] as the base learner. We establish its dynamic regret over arbitrary domains via a novel two-layer, mixability-based analysis.

The remainder of the paper is organized as follows. Section 1 introduces the problem setup, assumptions, and more related work. In Section 2, we present minimax-optimal regret bounds and fast-rate results for OPS. Section 3 develops efficient algorithms for OPS under the bounded-gradient assumption. Finally, Section 4 concludes the paper and discusses several future work.

# 1 Problem Setup and Related Work

This section introduces problem setup, assumptions, and additional related work for OPS.

## 1.1 Notation and Setup

Online portfolio selection proceeds over $T$ rounds of interaction between the learner and the market. At each round $t \in \{1, \dots, T\}$, the learner starts with wealth $S_{t-1}$ and distributes it across $d$ assets according to a probability vector $\mathbf{w}_t \in \Delta_d$, where $\Delta_d$ denotes the $(d-1)$-dimensional simplex. The market then reveals the non-negative price relative vector $\mathbf{x}_t \in \mathbb{R}_+^d$, where each component $x_{t,i} \geq 0$ represents the relative return of asset $i$. The learner's wealth is updated as $S_t = S_{t-1} \sum_{i=1}^d w_{t,i} \cdot x_{t,i}$. The learner's wealth becomes $S_T = S_0 \cdot \prod_{t=1}^T (\mathbf{w}_t^\top \mathbf{x}_t)$ after $T$ rounds. For OPS in non-stationary environments, our goal is to minimize the dynamic regret (1) with the log loss $\ell_t(\mathbf{w}) = -\ln(\mathbf{w}^\top \mathbf{x}_t)$:

$$\sum_{t=1}^T \ell_t(\mathbf{w}_t) - \sum_{t=1}^T \ell_t(\mathbf{u}_t) = -\ln \frac{\prod_{t=1}^T \mathbf{w}_t^\top \mathbf{x}_t}{\prod_{t=1}^T \mathbf{u}_t^\top \mathbf{x}_t},$$

which is equivalent to maximizing the logarithmic ratio of the learner's cumulative wealth to that of the time-varying investment strategy $\mathbf{u}_1, \dots, \mathbf{u}_T \in \Delta_d$.

## 1.2 Assumptions and Unbounded Gradient Issue

When learning with the log loss, a key challenge is that both the loss values and the norms of the corresponding gradients may be unbounded. Two classes of assumptions are typically imposed to ensure problem regularity and to facilitate the use of online learning techniques.

**Assumption 1** (No All-Zero Return)**.** *The return* $\mathbf{x}_t \in \mathbb{R}_+^d$ *satisfies* $\max_{i \in [d]} x_{t,i} > 0 \, \forall t \in [T]$.

**Assumption 2** (Bounded Gradient)**.** *The gradient of the loss is uniformly bounded by* $G$*, i.e.,*

$$\max_{t \in [T]} \|\nabla \ell_t(\mathbf{w})\|_\infty = \max_{t \in [T],\, i \in [d]} \frac{x_{t,i}}{\mathbf{w}^\top \mathbf{x}_t} \leq G, \forall \mathbf{w} \in \mathcal{W}.$$

Assumption 1 is arguably minimal for the OPS problem . Without it, the loss becomes unbounded for any investment strategy $\mathbf{w}$. This assumption is weaker than the one used in Cover [1991, Theorem 6.1] and is essentially the same as that in Cover and Ordentlich [1996]. Under this assumption alone, the Universal Portfolio method [Cover and Ordentlich, 1996] achieves the minimax-optimal static regret bound $\mathcal{O}(d \log T)$. This method requires integrating over the simplex to generate predictions, which can be implemented in polynomial time using log-concave sampling techniques [Kalai and Vempala, 2002].

Many subsequent works leverage tools from online convex optimization to obtain more efficient algorithms by imposing Assumption 2. This assumption is stronger than Assumption 1, but it can still be satisfied in several natural ways [Helmbold et al., 1998, Agarwal et al., 2006]. For example, one may

assume that the ratio of returns is bounded by $G$, that is, $\frac{\max_{i\in[d]} x_{t,i}}{\min_{i\in[d]} x_{t,i}} \leq G, \forall t \in [T]$, or alternatively restrict the set of portfolios to $\mathcal{W} = \{\mathbf{w} \in \Delta_d : \min_{i\in[d]} w_i \geq 1/G\}$. Under this assumption, the Exponential Gradient [Helmbold et al., 1998] algorithm achieves an $\mathcal{O}(G\sqrt{T})$ regret bound, while Online Newton Step [Agarwal et al., 2006] attains an $\mathcal{O}(dG \log T)$ bound with $\mathcal{O}(d^3)$ cost per round. However, in both cases the regret scales linearly with the gradient bound $G$, which can be reasonable in many situations but may still lead to loose guarantees when $G$ is large.

More recently, there has been a surge of interest in developing more efficient online learning algorithms under Assumption 1 [Orseau et al., 2017, Luo et al., 2018, Zimmert et al., 2022, Mhammedi and Rakhlin, 2022, Jézéquel et al., 2025]. Two Pareto-optimal results in terms of regret bounds and computational efficiency are VB-FTRL [Jézéquel et al., 2025], which achieves $\mathcal{O}(d \log T)$ regret with $\mathcal{O}(d^2 T)$ computational cost per round, and AdaMix+DONS [Mhammedi and Rakhlin, 2022], which attains an $\mathcal{O}(d^2 \log^5 T)$ regret bound with $\mathcal{O}(d^3 \log^2 T)$ per-round complexity.

### 1.3 Non-stationary OXO and Domain Constraint Issue

There are two lines of research that achieve an $\mathcal{O}(G \max\{\log T, T^{1/3} P_T^{2/3}\})$ dynamic regret for non-stationary OXO [Baby and Wang, 2021, Zhang et al., 2025]. Since these methods are primarily designed for general OXO purposes, their regret bounds typically scale with a bound on the gradient norm. However, setting aside the gradient-bound issue, these results still do not directly apply to OPS due to the restrictions imposed by domain constraints.

**Reduction-based Anlaysis.** An important research line for non-stationary OXO starts from Baby and Wang [2021] and is followed by Baby and Wang [2022a,b]. Under certain domain conditions, they provide a reduction from the interval regret bound [Hazan and Seshadhri, 2009], which guarantees a static regret bound on each interval, to a fast-rate dynamic regret bound. A key component of their analysis is a precise characterization of the optimal time-varying sequence $\{\mathbf{u}_t^*\}_{t=1}^T$ via KKT conditions and shows that $\{\mathbf{u}_t^*\}_{t=1}^T$ can be tracked by a piecewise-stationary sequence with $M = \mathcal{O}(T^{1/3} P_T^{2/3})$ switches. The initial work [Baby and Wang, 2021] requires improper learning, allowing the algorithm to predict in an extended box-constrained domain in order to obtain a sufficiently strong piecewise-stationary approximation. Later, Baby and Wang [2022a] show that proper learning can be achieved when the domain is exactly a box. This restriction to box constraints is intrinsic to the KKT-based analysis, as it only imposes coordinate-wise constraints on the optimal sequence. It remains unclear how to extend these analyses to the simplex or more general domains, which would introduce additional coupling constraints and complicate the analysis.

**Mixability-based Analysis** Recently, Zhang et al. [2025] showed that continuous exponential weights with a fixed-share update achieve fast-rate dynamic regret via mixability [Vovk, 1998], which lifts the analysis from pointwise predictors to distributional comparators. While this framework provides additional flexibility, its analysis relies on Gaussian comparators, which are incompatible with the simplex constraint. To enforce the domain constraint, the method requires an information projection onto a set of Gaussian mixture models with potentially infinitely many components, with bounded component means and variances, which makes the procedure computationally intractable. Our work builds on this framework and substantially improves its treatment of domain constraints. For OPS, we introduce a Dirichlet comparator that naturally handles unbounded losses and domain constraints, achieving fast rates without dependence on the large bound $G$. For general OXO, we derive an efficient projection scheme over individual Gaussian components via a refined two-layer analysis, leading to an efficient algorithm for non-stationary OXO. Further details on the mixability framework and technical analysis are deferred to Section 2.3 and Section 3.2.

## 2 Minimax Optimal and Fast Rate for OPS

This section establishes minimax-optimal and fast rates for the OPS problem under Assumption 1, which rules out the trivial case where the return vector $\mathbf{x}_t$ is identically zero. We first present a lower bound with a matching upper bound, and then discuss the case where fast rates are achievable.

## 2.1 Lower Bound

We establish an $\Omega(\max\{\log T, \sqrt{TP_T}\})$ lower bound for the OPS problem for all values of $P_T$.

**Theorem 1.** *Consider the OPS problem with 2 assets ($d = 2$). For any online algorithm and any $C \in [0, T]$, there exists a comparator sequence $u_1, \ldots, u_T \in [0, 1]$ and $\mathbf{x}_1, \ldots, \mathbf{x}_T \in \mathbb{R}^2_+$ such that*

$$P_T = \sum_{t=2}^{T} |u_t - u_{t-1}| \leq C \quad \textit{and} \quad \sum_{t=1}^{T} \ell_t(w_t) - \sum_{t=1}^{T} \ell_t(u_t) \geq \Omega(\max\{\log T, \sqrt{TC}\}),$$

This result reveals a strikingly different behavior of Cover's loss between stationary and relatively non-stationary environments. In the stationary environments, where $P_T = 0$, the Cover's loss enables $\mathcal{O}(\log T)$, an exponentially faster rate than the typical $\mathcal{O}(G\sqrt{T})$ rate for linear loss. By comparison, in non-stationary environments with $P_T = \omega(1)$, the minimax dynamic regret for OPS matches the $\Theta(G\sqrt{TP_T})$ rate for general convex loss in terms of the dependence on $T$. Any remaining room for improvement mainly lies in the dependence on the gradient norm bound $G$.

The proof of Theorem 1 is provided in Appendix B.1 via a reduction to sequential probability assignment (SPA). Briefly, we consider the two-asset Kelly market with returns restricted to $\mathbf{x}_t \in \{\mathbf{e}_1, \mathbf{e}_2\}$, where $\mathbf{e}_i$ is the $i$-th standard basis vector. Under this restriction, OPS reduces to SPA with binary outcomes [Cesa-Bianchi and Lugosi, 2006, Chapter 9.1], and its minimax dynamic regret coincides with the minimax regret of SPA over total-variation classes. We construct a piecewise-stationary environment by partitioning the horizon into $K = \mathcal{O}(\sqrt{CT})$ blocks of length approximately $1/\varepsilon$ with $\varepsilon = \sqrt{C/T}$, and on each block randomly choosing between two Bernoulli models with parameters 0 and $\varepsilon$, both close to the boundary. We show that any algorithm will incur a constant regret on each block, which yields an $\Omega(\sqrt{CT})$ lower bound.

We note that Bilodeau et al. [2020], Jia et al. [2025] establishes an $\Omega(\sqrt{T})$ lower bound for SPA with a 1-Lipschitz function class in an adversarial contextual setting. Although the 1-Lipschitz class is contained in the total-variation class, that result does not directly apply here because our problem is non-contextual, where the minimax regret could be smaller. Moreover, our lower bound holds uniformly for any path length budget $C$, rather than only for a fixed instance such as $C = 1$.

## 2.2 Minimax Optimal Rate

There are no existing results establishing a $G$-free dynamic regret for OPS. We show that the minimax optimal rate can be achieved via a simple reduction-based argument up to logarithimic factors.

**Lemma 1.** *For the OPS problem, assume there exists an online algorithm $\mathcal{A}$ that attains the following interval regret guarantee, such that, for any interval $\mathcal{I} \subseteq [T]$,*

$$\sum_{t \in \mathcal{I}} \ell_t(\mathbf{w}_t) - \min_{\mathbf{w} \in \Delta_d} \sum_{t \in \mathcal{I}} \ell_t(\mathbf{w}) \leq B(T), \tag{2}$$

*with a certain function $B : \mathbb{N} \to \mathbb{R}_+$. Then, the dynamic regret of the algorithm $\mathcal{A}$ is bounded by*

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) \leq 2B(T) + 2\sqrt{B(T)TP_T^{(1)}},$$

*where $P_T^{(1)} = \sum_{t=2}^{T} \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_1$ is the path length defined in terms of the $L_1$ norm.*

Lemma 1 provides a black-box reduction from interval regret [Hazan and Seshadhri, 2009] to dynamic regret for the OPS problem. A key advantage of this guarantee is that it does not require a bounded gradient norm, provided that the algorithms are chosen appropriately. For OPS, such algorithms can be constructed with $B(T) = \mathcal{O}(d \log T)$. For example, one may run the FLH [Hazan and Seshadhri, 2009] algorithm with Universal Portfolio [Cover, 1991] or VB-FTRL [Jézéquel et al., 2025] as the base learner. In this case, the resulting dynamic regret matches the optimal rate up to logarithmic factors. The proof of Lemma 1 is given in Appendix B.2. It uses a bin-partition argument to show that the cumulative loss of any comparator sequence can be approximated by that of a piecewise-stationary sequence with $M = \mathcal{O}(\sqrt{TP_T})$ switches. One can then use an algorithm with interval regret bounds to track this piecewise-stationary sequence at a low additional cost.

**Algorithm 1** Universal Dynamic Portfolio

**Require:** fixed-share parameter $\mu_t = 1/t$.

Initialize $\widetilde{P}_1 = P_1 = \mathrm{Dir}(\boldsymbol{\alpha}_1)$ as a Dirichlet distribution with parameters $\boldsymbol{\alpha}_1 = \mathbf{1}$.

**for** $t = 1, 2, \ldots, T$ **do**

The learner submits the prediction $\mathbf{w}_t = \mathbb{E}_{\mathbf{u}\sim P_t}[\mathbf{u}]$ and then observes $\mathbf{x}_t \in \mathbb{R}^d_+$.

The learner updates the distributions by

$$\widetilde{P}_{t+1}(\mathbf{u}) \propto P_t(\mathbf{u}) \exp(-\ell_t(\mathbf{u})), \forall \mathbf{u} \in \Delta_d; \tag{3}$$

$$P_{t+1}(\mathbf{u}) = (1-\mu_{t+1})\widetilde{P}_{t+1}(\mathbf{u}) + \mu_{t+1}\mathrm{Dir}(\boldsymbol{\alpha}_1). \tag{4}$$

**end for**

We note that reduction-based arguments are widely used in the dynamic regret minimization literature [Cutkosky, 2020, Baby and Wang, 2021]. The main distinction in our setting is that the loss functions in OPS do not admit a uniform Lipschitz constant. In contrast, existing analyses in OCO typically rely on a bounded gradient norm to relate the instantaneous loss difference to the path length, e.g., $\ell_t(\mathbf{u}) - \ell_t(\mathbf{v}) \le G\|\mathbf{u}-\mathbf{v}\|$. Nevertheless, we show this issue can be overcome by a direct treatment of the log loss. Besides, our analysis does not need the KKT-based arguments employed in Baby and Wang [2021], since our objective is limited to achieving an $\mathcal{O}(\sqrt{TP_T})$ bound, which is known to be minimax optimal by the lower bound in Theorem 1.

## 2.3 Fast Rate Guarantee under Margin Condition

In the proof of Theorem 1, we construct a hard instance based on a piecewise-stationary stochastic environment to establish the lower bound. A key feature of this construction is that the optimal comparator sequence lies close to the boundary of the domain, which forces any algorithm to incur constant regret on each stationary interval. In contrast, when the comparator sequence stays strictly in the interior of the domain with a positive margin, this per-segment constant regret can be avoided in our example, which in turn suggests that faster rates may be achievable.

Building on this observation, we propose Algorithm 1 (Universal Dynamic Portfolios) for achieving fast rates under the margin condition, which can be viewed as Cover's Universal Portfolio equipped with a fixed-share update. The dynamic regret is presented below.

**Theorem 2.** *Under Assumption 1, Algorithm 1 with $\mu_t = 1/t$ ensures*

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}\left(d\left(\alpha^{-\frac{1}{3}} T^{\frac{1}{3}} P_T^{\frac{2}{3}} (\ln(Td))^{\frac{2}{3}} + \ln(dT/\alpha)\right)\right),$$

*for any sequence $\mathbf{u}_1, \ldots, \mathbf{u}_T \in \Delta_d$ that $u_{t,i} \ge \alpha \in (0, 1/d]$ for all $i \in [d]$ and $t \in [T]$, where $P_T = \sum_{t=2}^T \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$ is the path length of the comparator sequence.*

We emphasize that our method is free of parameter tuning: its guarantee automatically adapts to any comparator, any margin $\alpha > 0$, and any time horizon $T$ without requiring their prior knowledge.

**Recover Minimax Optimal Rate.** In addition, we show that the margin-based bound directly implies the minimax-optimal guarantee for any comparator. This is achieved by constructing a smoothed counterpart $\widetilde{\mathbf{u}}_t = (1-\beta)\mathbf{u}_t + (\beta/d)\mathbf{1}$ for each $\mathbf{u}_t \in \Delta_d$. As shown in Appendix C.3, the dynamic regret admits the decomposition $\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) \le \text{D-Reg}(\{\widetilde{\mathbf{u}}_t\}_{t=1}^T) + T\ln(1/(1-\beta))$, which yields the minimax bound up to a logarithmic factor by appropriately balancing $\beta$.

**Corollary 1.** *Assume $T \ge 2d$. Let $\mathcal{A}$ be an algorithm such that, for any comparator sequence $\mathbf{u}_1 \ldots, \mathbf{u}_T \in \Delta_d$ satisfying $u_{t,i} \ge \alpha$ for all $i \in [d]$ and $t \in [T]$, it guarantees*

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) \le d\,B\left(\alpha^{-1/3} T^{1/3} P_T^{2/3} + \ln\frac{T}{\alpha}\right),$$

*where $B$ hides logarithmic factors in $d$ and $T$ and absolute constants. Then $\mathcal{A}$ also guarantees, for any comparator sequence $\mathbf{u}_1 \ldots, \mathbf{u}_T \in \Delta_d$,*

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}\left(dB\left(\ln(dT) + \sqrt{TP_T}\right)\right).$$

**Discussion with the Reduction-based Method.** Since we now compete against a comparator sequence $\{\mathbf{u}_t\}_{t=1}^T$ lying in the relative interior of the simplex, which permits improper learning in the OPS setting, it is natural to ask whether a reduction-based approach, such as the one presented in Lemma 1, can yield fast rates. However, extending this argument appears nontrivial. In particular, for the reduction-based method to achieve fast rates, the comparator sequence should be approximated by a piecewise-stationary sequence with $\mathcal{O}(T^{1/3}P_T^{2/3})$ switches, which is a stronger requirement than what we establish in the proof of Lemma 1. Under box-constrained domains, Baby and Wang [2021] show that such an approximation can be obtained via a careful analysis of the KKT conditions of the optimal comparators. However, in the OPS setting, even when the comparators lie in the relative interior of the domain, the simplex constraint still appears in the KKT conditions, making the analysis more delicate.

***Proof Sketch*** Here, we adopt an alternative approach to obtain fast rates based on the notion of mixability, which is first used to analyze the prediction with expert advice problem [Vovk, 1998] and found useful for achieving fast rate for both stochastic learning and online learning [Vovk, 2001, van Erven et al., 2015, Foster et al., 2018]. Recently, Zhang et al. [2025] used this notion to obtain fast-rate dynamic regret bounds. However, their analysis does not directly extend to OPS under Assumption 1. We provide a more detailed discussion after briefly sketching the proof of Theorem 2 below. The complete proof is provided in Appendix C.2.

The starting point of our analysis is that Cover's loss for OPS is 1-mixable over the simplex $\Delta_d$ for any $\mathbf{x}_t \in \mathbb{R}_+^d$, in the sense that for any distribution $P_t$ over $\Delta_d$ and $\mathbf{w}_t = \mathbb{E}_{\mathbf{u}\sim P_t}[\mathbf{u}]$, we have $\ell_t(\mathbf{w}_t) = -\ln\left(\mathbb{E}_{\mathbf{u}\sim P_t}[\exp(-\ell_t(\mathbf{u}))]\right)$, which implies the following variational identity.

**Lemma 2.** *For any distribution $Q_t$ with $\operatorname{supp}(Q_t) \subseteq \operatorname{supp}(P_t)$, it holds that*

$$\ell_t(\mathbf{w}_t) = \mathbb{E}_{\mathbf{u}\sim Q_t}[\ell_t(\mathbf{u})] + \mathrm{KL}(Q_t \,\|\, P_t) - \mathrm{KL}(Q_t \,\|\, P'_{t+1}),$$

*where the distribution $P'_{t+1}$ is defined by $P'_{t+1}(\mathbf{u}) \propto P_t(\mathbf{u})\exp(-\ell_t(\mathbf{u}))$ for all $\mathbf{u} \in \operatorname{supp}(P_t)$.*

By the update rules (3) and (4), we have $P_{t+1} = (1-\mu_{t+1})P'_{t+1} + \mu_{t+1}\mathrm{Dir}(\mathbf{1})$. Telescoping over $T$ iterations and upper bound the $\mu_t = 1/t$ fraction difference between $P_{t+1}$ and $P'_{t+1}$ yields

$$\sum_{t=1}^T \ell_t(\mathbf{w}_t) \lesssim \sum_{t=1}^T \mathbb{E}_{\mathbf{u}\sim Q_t}[\ell_t(\mathbf{u})] + \sum_{t=2}^T \int_{\mathbf{u}\in\Delta_d} (Q_t(\mathbf{u}) - Q_{t-1}(\mathbf{u})) \ln \frac{1}{P_t(\mathbf{u})}\,\mathrm{d}\mathbf{u},$$

where $\lesssim$ hides additive logarithmic factors in $T$. In our analysis, to accommodate the simplex constraint, we choose the comparator $Q_t = \mathrm{Dir}(1+\gamma\mathbf{u}_t)$ as a Dirichlet distribution, where $\gamma > 0$ is a parameter to be tuned and $\mathbf{u}_t$ is the comparator sequence.

The use of Dirichlet distributions also effectively resolves the unboundedness of the log loss, as it assigns low density near the boundary of the domain. By a careful analysis exploiting the structure of the Dirichlet distribution, we can show that

$$\sum_{t=1}^T \mathbb{E}_{Q_t}[\ell_t(\mathbf{u})] \le \sum_{t=1}^T \ell_t(\mathbf{u}_t) + \frac{dT}{\gamma} \quad \text{and} \quad \sum_{t=2}^T \int_{\mathbf{u}\in\Delta_d} (Q_t(\mathbf{u}) - Q_{t-1}(\mathbf{u})) \ln \frac{1}{P_t(\mathbf{u})}\mathrm{d}\mathbf{u} \lesssim d\sqrt{\frac{\gamma}{\alpha}}P_T,$$

where the first inequality follows from reparameterizing the Dirichlet distribution by several Gamma distributions and does not impose any requirement on gradient boundedness. Combining these bounds and choosing parameters $\gamma = \Theta(\alpha^{1/3}T^{\frac{2}{3}}P_T^{-2/3})$ yields the desired result. □

In prior work [Zhang et al., 2025], the comparator distribution was chosen to be Gaussian for analytical convenience. However, since the definition requires the domain to be $\mathbb{R}^d$, this choice is not applicable to constrained domains. To handle general constraints, Zhang et al. [2025] proposed learning with a quadratic surrogate loss that extends the domain to $\mathbb{R}^d$, which requires a bounded gradient. Besides, when updating with the surrogate loss, the mean of the learned distribution $P_{t+1}$ may lie outside the constraint set. To address this issue, they project $P_t$ onto a family of Gaussian mixture models with possibly infinitely many components and bounded component means and variances, which is generally computationally intractable. By contrast, we adopt a Dirichlet comparator to naturally accommodate the simplex constraint and the unbounded loss.

**Algorithm 2** Follow-the-Leading-History

**Require:** Loss parameter $\eta = \frac{1}{5}\min\{1/(2G), 1\}$ and fixed-share parameter $\mu_t = 1/t$.

1: Initialize $P_1 = \mathcal{N}(\mathbf{u}_0, I_d)$ as a Gaussian distribution with mean $\mathbf{u}_0 \in \Delta_d$.

2: Initialize a pool with base-learners $\mathcal{H}_1 = \{\mathcal{B}_1\}$, where $\mathcal{B}_1$ is the initial base-learner with the distribution $P_{1,1} = P_1$ and weight $p_{1,1} = 1$.

3: **for** $t = 1, 2, \ldots, T$ **do**

4: The learner submits the prediction $\mathbf{w}_t = \mathbb{E}_{\mathbf{u}\sim P_t}[\mathbf{u}]$ and then obtains the surrogate loss (8).

5: For all base-learners $\mathcal{B}_i \in \mathcal{H}_t$, we update by

$$\begin{cases} P'_{t+1,i}(\mathbf{u}) \propto P_{t,i} \cdot e^{-\eta\widetilde{\ell}_t(\mathbf{u})}, \ \forall \mathbf{u} \in \mathbb{R}^d \\ P_{t+1,i}(\mathbf{u}) = \arg\min_{Q\in\mathscr{W}} \mathrm{KL}\big(Q \,\|\, P'_{t+1,i}\big), \end{cases} \tag{5}$$

where $\mathscr{W} = \{Q : \mathbb{E}_{\mathbf{u}\in Q}[\mathbf{u}] \in \Delta_d\}$ is the set of distributions whose means lie in $\Delta_d$.

6: Update the weight for each base-learner $\mathcal{B}_i \in \mathcal{H}_t$ by

$$\widetilde{p}_{t+1,i} \propto p_{t,i} \cdot \mathbb{E}_{\mathbf{u}\sim P_{t,i}}[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))]. \tag{6}$$

7: Initialize a new base-learner $\mathcal{B}_{t+1}$ with the distribution $P_{t+1,t+1} = \mathcal{N}(\mathbf{u}_0, I_d)$ and update the weight for existing base-learner by

$$p_{t+1,i} = \begin{cases} (1-\mu_{t+1}) \cdot \widetilde{p}_{t+1,i} \ \text{ for } \mathcal{B}_i \in \mathcal{H}_t \\ \mu_{t+1} \text{ for } \ \mathcal{B}_i = \mathcal{B}_{t+1}. \end{cases} \tag{7}$$

8: Update the pool $\mathcal{H}_{t+1} = \mathcal{H}_t \cup \{\mathcal{B}_{t+1}\}$ and obtain $P_{t+1}(\mathbf{u}) = \sum_{\mathcal{B}_i\in\mathcal{H}_{t+1}} p_{t+1,i} \cdot P_{t+1,i}(\mathbf{u})$.

9: **end for**

### 2.4 Equivalent Implementation and Interval Regret Guarantee

Following the same arguments in Adamskiy et al. [2016], Zhang et al. [2025], one can show Algorithm 1 is equivalent to running the FLH algorithm with Universal Portfolio as the base learner. Therefore, our method naturally enjoys the interval regret guarantee, which also implies a bound on the switching regret, where the comparator sequence is allowed to switch up to $K$ times.

**Proposition 1.** *Under the same condition as Theorem 2, Algorithm 1 ensures*

$$\sum_{t\in\mathcal{I}} \ell_t(\mathbf{w}_t) - \sum_{t\in\mathcal{I}} \ell_t(\mathbf{u}) \le \mathcal{O}(d\log T),$$

*for any interval $\mathcal{I} = [r, s] \subseteq [T]$ and any comparator $\mathbf{u} \in \Delta_d$. Furthermore, let $K = \sum_{t=2}^{T} \mathbb{1}\{\mathbf{u}_t \neq \mathbf{u}_{t-1}\}$, Algorithm 1 ensures* $\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}(d(K+1)\log T)$.

## 3 Towards Fast Rates and Efficiency for General OXO

While Algorithm 1 achieves minimax-optimal and fast rates for OPS, it still need to compute an integral over the simplex. In this section, we develop a computationally efficient alternative under the bounded-gradient assumption (Assumption 2), mirroring the development in static regret that from Universal Universal Portfolio (under Assumption 1) to ONS (with Assumption 2).

Interestingly, our algorithm can be viewed as a variant of the FLH method [Hazan and Seshadhri, 2009] with ONS [Hazan et al., 2007] as the base learner. While FLH is a well-known method for non-stationary online learning, its performance is typically analyzed in terms of interval regret. Here, we show that FLH-ONS achieves fast-rate dynamic regret via a novel two-layer, mixability-based analysis. Our analysis does not exploit any structure beyond exp-concavity and bounded gradients and domains, and therefore applies to general OXO with arbitrary convex domains.

### 3.1 Proposed Method

A standard technique to ensure efficient OXO is learning with the surrogate loss:

$$\widetilde{\ell}_t(\mathbf{w}) \;=\; \mathbf{g}_t^\top(\mathbf{w}-\mathbf{w}_t) + \eta\,\|\mathbf{w}-\mathbf{w}_t\|^2_{\mathbf{g}_t\mathbf{g}_t^\top}, \tag{8}$$

where $\mathbf{g}_t = \nabla \ell_t(\mathbf{w}_t)$ is the gradient of the loss function. For $\kappa$-exp-concave loss function over domain $\mathcal{W}$, Hazan [2016, Lemma 4.3] shows that one can upper bound the regret in terms of original loss by that of the surrogate loss, i.e., $\ell_t(\mathbf{w}_t) - \ell_t(\mathbf{u}_t) \;\leq\; \widetilde{\ell}_t(\mathbf{w}_t) - \widetilde{\ell}_t(\mathbf{u}_t)$ for any $\mathbf{u}_t \in \mathcal{W}$ when we set $\eta \leq \frac{1}{4}\min\left\{\left(\max_{\mathbf{w}\in\mathcal{W}} |\mathbf{g}_t^\top(\mathbf{w}-\mathbf{w}_t)|\right)^{-1}, \kappa^{-1}\right\}$. Since Cover's loss is 1-exp-concave, it is sufficient to set $\eta \leq \frac{1}{4}\min\{1/(2G), 1\}$ under Assumption 2.

Our algorithm is summarized in Algorithm 2. It is essentially the FLH algorithm [Hazan and Seshadhri, 2009] with a two-layer design, consisting of a meta-learner that aggregates the predictions of multiple base learners. At each iteration $t = i$, we initialize a new base learner $\mathcal{B}_i$ and add it to the expert pool $\mathcal{H}_t$. The base-learner is initialized with a Gaussian distribution $P_{t,i} = \mathcal{N}(\mathbf{u}_0, I_d)$, where $\mathbf{u}_0 \in \Delta_d$ can be any point in the interior of the feasible domain. The distribution of each base learner $\mathcal{B}_i$ is updated using exponential weights with respect to the surrogate loss. Since the surrogate loss $\widetilde{\ell}_t$ is quadratic, van der Hoeven et al. [2018, Theorem 5] show that the resulting distribution $P_{t,i} = \mathcal{N}(\mathbf{w}_{t,i}, H_{t,i}^{-1})$ remains Gaussian, with its mean and covariance updated via an ONS-type rule. An explicit update formula is provided in (26) in Appendix D. In addition, for each base learner $\mathcal{B}_i \in \mathcal{H}_t$, we maintain a meta-learner that updates its weight $p_{t,i}$ based on its performance, as shown in (6). The resulting aggregated $P_t$ distribution is therefore a mixture of Gaussians. The prediction is given by $\mathbf{w}_t = \mathbb{E}_{\mathbf{u}\sim P_t}[\mathbf{u}] = \sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\,\mathbf{w}_{t,i}$.

One slight difference between Algorithm 2 and FLH is that (6) updates the weights using the expectation term $\mathbb{E}_{\mathbf{u}\sim P_{t,i}}[e^{-\eta\widetilde{\ell}_t(\mathbf{u})}]$ instead of the classical one $p_{t+1,i} \propto p_{t,i}e^{-\eta\widetilde{\ell}_t(\mathbf{w}_{t,i})}$. This difference is important for our analysis, as it allows us to align Algorithm 2 with exponential-weights updates over distributions. We also note that (6) admits a closed-form expression, since $P_{t,i}$ is Gaussian and $\widetilde{\ell}_t$ is a quadratic function. We have the following guarantee, whose proof is provided in Appendix D.

**Theorem 3.** *Under Assumption 2, Algorithm 2 with $\mu_t = 1/t$ and $\eta = \frac{1}{5}\min\{1/(2G), 1\}$ ensures*

$$\text{D-Reg}_T(\{\mathbf{u}_t\}_{t=1}^T) \leq \mathcal{O}\left(dG\left(\ln(dT) + T^{\frac{1}{3}}P_T^{\frac{2}{3}}(\ln(Td))^{\frac{2}{3}}\right)\right),$$

*for any sequence $\mathbf{u}_1, \ldots, \mathbf{u}_T \in \Delta_d$, where $P_T = \sum_{t=2}^T \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$ is the path length.*

### 3.2 Two-layer Mixability-based Analysis

This part sketches the proof of Theorem 3. The FLH admits a classical method, yet much of the existing understanding focuses on its interval regret guarantees. While interval regret bounds can sometimes be leveraged to obtain dynamic regret guarantees, such reductions are typically nontrivial. For instance, the result of Baby and Wang [2021] holds only for box-constraint domain. In this part, we present a novel analysis on the FLH-type method through the lens of mixability. This reveals that FLH can in fact achieve fast-rate dynamic regret over arbitrary convex domains.

**Limitations of Previous Attempts.** Zhang et al. [2025] showed that although the surrogate loss is not mixable in general, mixability-based analysis still applies when the mean of each component $P_{t,i}$ in the Gaussian mixture $P_t$ lies in the feasible domain, yielding the variational formulation:

$$\widetilde{\ell}_t(\mathbf{w}_t) \leq \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \mathrm{KL}(Q_t \,\|\, P_t) - \mathrm{KL}(Q_t \,\|\, P'_{t+1}), \tag{9}$$

where $P'_{t+1}(\mathbf{u}) \propto P_t(\mathbf{u})\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))$, which remains a Gaussian mixture model. However, the exponential weights update does not guarantee that the component means of $P'_{t+1}$ stay within the decision domain, and thus one cannot set $P_{t+1} = P'_{t+1}$ to telescope the KL terms over $T$ rounds. To overcome this issue, Zhang et al. [2025] project $P'_{t+1}$ onto a set of Gaussian mixtures with bounded component means and covariances, which ensures a KL–Pythagorean inequality such that $\mathrm{KL}\left(Q \,\|\, P_{t+1}\right) \leq \mathrm{KL}\left(Q \,\|\, P'_{t+1}\right)$ for any $Q$ with $\mathbb{E}_Q[\mathbf{w}] \in \Delta_d$. This allows telescoping and yields

$$\sum_{t=1}^T \widetilde{\ell}_t(\mathbf{w}_t) \lesssim \sum_{t=1}^T \mathbb{E}_{Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \sum_{t=2}^T \big(\mathrm{KL}(Q_t\|P_t) - \mathrm{KL}(Q_{t-1}\|P_t)\big). \tag{10}$$

By choosing $Q_t = \mathcal{N}(\mathbf{u}_t, \sigma^2 I_d)$ and selecting $\sigma$ properly, the above inequality leads to the desired fast rate bound. This analysis does not apply to Algorithm 2, since our method projects each component $P'_{t+1,i}$ into the domain separately to gain computational efficiency, and such componentwise projections do not satisfy the required KL–Pythagorean guarantee for $P_{t+1}$ and $P'_{t+1}$.

**Our Analysis.** We overcome the projection issue by exploiting the two-layer structure, rather than applying mixability at the level of the aggregated distribution. Specifically, for any distribution $Q$ with $\mathbb{E}_Q[\mathbf{u}] \in \Delta_d$ and $\mathbf{q} \in \Delta_{|\mathcal{H}_t|}$, we have

$$\begin{aligned}\widetilde{\ell}_t(\mathbf{w}_t) \lesssim \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\Bigg(\sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\mathrm{KL}\left(Q_t \,\|\, P_{t,i}\right) + \mathrm{KL}\left(\mathbf{q}_t \,\|\, \mathbf{p}_t\right)\Bigg) \\ - \frac{1}{\eta}\Bigg(\sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\mathrm{KL}\left(Q_t \,\|\, P_{t+1,i}\right) + \mathrm{KL}\left(\mathbf{q}_t \,\|\, \widetilde{\mathbf{p}}_{t+1}\right)\Bigg).\end{aligned} \tag{11}$$

The above variational-form bound can be viewed as a two-layer counterpart of (9), providing the flexibility to choose the comparators $\mathbf{q}_t$ and $Q_t$ for the meta-learner and the base-learner, respectively. We note that componentwise projections can be performed safely under (11), since the KL divergence is defined in terms of the individual distributions rather than the aggregated one.

The next question is how to choose $\mathbf{q}_t$ and $Q_t$. To make the bound as tight as possible, we choose $\mathbf{q}_t = \arg\min_{\mathbf{q}\in\Delta_{|\mathcal{H}_t|}} \sum_{\mathcal{B}_i\in\mathcal{H}_t} q_i\mathrm{KL}\left(Q_t \,\|\, P_{t,i}\right) + \mathrm{KL}\left(\mathbf{q} \,\|\, \mathbf{p}_t\right)$, whose optimal value attains a closed-form formula $V_t(Q_t) = -\ln\left(\mathbb{E}_{\mathbf{p}_t}[\exp(-\mathrm{KL}\left(Q_t \,\|\, P_{t,i}\right))]\right)$. After a sequence of algebraic manipulations, the bound admits a telescoping structure as in (10).

$$\sum_{t=1}^{T}\widetilde{\ell}_t(\mathbf{w}_t) \lesssim \sum_{t=1}^{T}\mathbb{E}_{Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\sum_{t=2}^{T}\left(V_t(Q_t) - V_t(Q_{t-1})\right), \tag{12}$$

By specifying $Q_t = \mathcal{N}(\mathbf{u}_t, \sigma^2 I_d)$ as a Gaussian distribution, we can further show that

$$\mathbb{E}_{Q_t}[\widetilde{\ell}_t(\mathbf{u})] \le \widetilde{\ell}_t(\mathbf{u}_t) + d\sigma^2 GT \quad \text{and} \quad \sum_{t=2}^{T}(V_t(Q_t) - V_t(Q_{t-1})) \lesssim \frac{dGP_T}{\sigma}.$$

The first inequality follows from the quadratic formulation of the surrogate loss function, and the second inequality is obtained by showing that the gradient of $V_t(Q_t)$ with respect to $\mathbf{u}_t$ can be upper bound by $\mathcal{O}(d\log T/\sigma)$. We can obtain the desired bound by setting $\sigma = \Theta\big(P_T^{1/3}T^{-1/3}\big)$.

# 4 Conclusion

This work provides several upper bounds and lower bound for understanding the behavior of Cover's loss for non-stationary OPS. Under nozero return assumption, we show an $\Theta(\max\{\log T, \sqrt{TP_T}\})$ minimax-optimal rate and $\mathcal{O}(\max\{\log T, \alpha^{-1/3}T^{1/3}P_T^{2/3}\})$ fast rate. For bounded-gradient setting, we propose a computationally efficient algorithm that achieves $\mathcal{O}(G\max\{\log T, T^{1/3}P_T^{2/3}\})$ bound, which is applicable to general OXO over arbitrary convex domains.

There are two interesting directions for further exploration. *(i)* Dimension dependence: This work focuses on optimality in terms of $T$ and $P_T$, while leaving the dependence on $d$ implicit. Our lower bound argument holds for $d = 2$, and it is interesting to further characterize the optimal dependence on the dimension for the dynamic regret of OPS. *(ii) More efficient methods:* In the nonzero-return setting, our fast-rate results rely on exponential-weights updates that require integration, leaving open the design of more efficient algorithms with comparable fast-rate dynamic regret bounds. In the bounded-gradient setting, our algorithm admits an FLH-ONS structure but still requires maintaining $\mathcal{O}(T)$ base learners. It is interesting to study whether geometric covering techniques [Daniely et al., 2015] can be applied to reduce the computational cost to $\mathcal{O}(\log T)$.

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# A Properties of the Dirichlet Distribution

In this section, we present some useful properties of the Dirichlet distribution that will be used in our analysis.

**Definition 1** (Dirichlet Distribution)**.** *A random vector* $\mathbf{w} \in \Delta_d$ *is said to follow a Dirichlet distribution with parameter vector* $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d) \in \mathbb{R}_+^d$*, denoted by* $\mathbf{w} \sim \mathrm{Dir}(\boldsymbol{\alpha})$*, if its probability density function is*

$$p(\mathbf{w} \mid \boldsymbol{\alpha}) = \frac{\Gamma\left(\sum_{i=1}^d \alpha_i\right)}{\prod_{i=1}^d \Gamma(\alpha_i)} \prod_{i=1}^d w_i^{\alpha_i - 1}, \qquad \mathbf{w} \in \Delta_d.$$

*where* $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} \mathrm{d}t$ *is the Gamma function.*

**Proposition 2** (Properties of Dirichlet Distribution)**.** *Let* $\mathbf{w} \sim \mathrm{Dir}(\boldsymbol{\alpha})$ *and* $\alpha_0 = \sum_{i=1}^d \alpha_i$*. Then,*

- *The mean of the random vector is given by* $\mathbb{E}_{\mathbf{w} \sim \mathrm{Dir}(\boldsymbol{\alpha})}[\mathbf{w}] = \boldsymbol{\alpha}/\alpha_0$*.*
- *The covariance matrix is* $\mathrm{Cov}[\mathbf{w}] = \frac{1}{\alpha_0^2(\alpha_0+1)}(\alpha_0 \mathrm{diag}(\boldsymbol{\alpha}) - \boldsymbol{\alpha}\boldsymbol{\alpha}^\top)$*.*
- *Dirichlet distribution belongs to the exponential family with natural parameter* $\boldsymbol{\eta} = \boldsymbol{\alpha} - \mathbf{1}$ *and log-partition function* $A(\boldsymbol{\eta}) = \sum_{i=1}^d \ln \Gamma(\alpha_i) - \ln \Gamma(\alpha_0)$*.*
- *The KL divergence between two Dirichlet distributions* $\mathrm{Dir}(\boldsymbol{\alpha})$ *and* $\mathrm{Dir}(\boldsymbol{\beta})$ *is given by*

$$\mathrm{KL}(\mathrm{Dir}(\boldsymbol{\alpha}) \| \mathrm{Dir}(\boldsymbol{\beta})) = \ln \frac{\Gamma(\alpha_0)}{\Gamma(\beta_0)} - \sum_{i=1}^d \ln \frac{\Gamma(\alpha_i)}{\Gamma(\beta_i)} + \sum_{i=1}^d (\alpha_i - \beta_i)\left(\psi(\alpha_i) - \psi(\alpha_0)\right),$$

  *where* $\psi(\alpha) = \frac{d}{d\alpha} \ln \Gamma(\alpha)$ *is the digamma function.*
- *The differential Shannon entropy of* $\mathrm{Dir}(\boldsymbol{\alpha})$ *is*

$$H(\mathrm{Dir}(\boldsymbol{\alpha})) = \ln \left( \frac{\prod_{i=1}^d \Gamma(\alpha_i)}{\Gamma(\alpha_0)} \right) + (\alpha_0 - d)\psi(\alpha_0) - \sum_{i=1}^d (\alpha_i - 1)\psi(\alpha_i).$$

**Proposition 3** (Properties of Gamma Function)**.** *Let* $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} \mathrm{d}t$ *be the Gamma function and* $\psi(\alpha) = \frac{d}{da} \ln \Gamma(\alpha)$ *be the digamma function. Then, for any* $\alpha > 0$*, we have*

- $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$ *and* $\psi(\alpha + 1) = \psi(\alpha) + 1/\alpha$*.*
- $\ln(\alpha) - \frac{1}{\alpha} \leq \psi(\alpha) \leq \ln(\alpha) - \frac{1}{2\alpha}$*.*

**Lemma 3.** *Let* $P = \mathrm{Dir}(\mathbf{1} + \gamma \mathbf{u})$*,* $Q = \mathrm{Dir}(\mathbf{1} + \gamma \mathbf{e}_i)$ *and* $P_1 = \mathrm{Dir}(\mathbf{1})$*. Then,* $\mathrm{KL}(P \,\|\, P_1) \leq \mathrm{KL}\left(Q \,\|\, P_1\right)$ *for any* $\mathbf{u} \in \Delta_d$*,* $i \in [d]$ *and* $\gamma > 0$*.*

*Proof of Lemma 3.* By definition of the KL divergence between two Dirichlet distributions, we have

$$\mathrm{KL}\left(P \,\|\, P_1\right) = \ln \frac{\Gamma(d+\gamma)}{\Gamma(d)} - \gamma\psi(d+\gamma) + \sum_{i=1}^d \left(\gamma u_i \psi(1+\gamma u_i) - \ln \Gamma(1+\gamma u_i)\right).$$

Let $G(\mathbf{u}) = \sum_{i=1}^d g(u_i)$ where $g(u) = \gamma u \psi(1+\gamma u) - \ln \Gamma(1+\gamma u)$. According to Lemma 5, one can show that $G(\mathbf{u})$ is a convex function over the simplex. Then, we have $G(\mathbf{u}) = G(\sum_{j=1}^d u_j \mathbf{e}_j) \leq \sum_{j=1}^d u_j G(\mathbf{e}_j) \leq G(\mathbf{e}_i)$ for any $i \in [d]$, where the last inequality holds because $G(\mathbf{u})$ is invariant under permutations of the coordinates. Then, we complete the proof by showing

$$\mathrm{KL}\left(P \,\|\, P_1\right) = \ln \frac{\Gamma(d+\gamma)}{\Gamma(d)} - \gamma\psi(d+\gamma) + G(\mathbf{u}) \leq \ln \frac{\Gamma(d+\gamma)}{\Gamma(d)} - \gamma\psi(d+\gamma) + G(\mathbf{e}_i) = \mathrm{KL}\left(Q \,\|\, P_1\right).$$

□

# B Omitted Proofs for Section 2.1 and Section 2.2

## B.1 Proof of Theorem 1

*Proof of Theorem 1.* We focus on the minimax regret for the 2-assets OPS problem:

$$\mathcal{W}_T(\mathcal{U}_C) = \inf_{w_1,\dots,w_T\in[0,1]} \sup_{\mathbf{x}_1,\dots,\mathbf{x}_T\in\mathbb{R}^2_+} \sup_{u_{1:T}\in\mathcal{U}_C} \left(\sum_{t=1}^T \ell_t(w_t) - \sum_{t=1}^T \ell_t(u_t)\right),$$

where $w_t$ is the algorithm's prediction based only on past observations $\mathbf{x}_1,\dots,\mathbf{x}_{t-1}$. We write $u_{1:T} = (u_1,\dots,u_T)$ and define $\mathcal{U}_C = \{u_{1:T}\in[0,1]^T \mid \sum_{t=2}^T |u_t - u_{t-1}| \le C\}$ is the set of all comparator sequence whose path length is smaller than $C$.

**Reduction to SPA.** For any $C\in[0,T]$, we derive a lower bound on the value $\mathcal{W}_T(\mathcal{U}_C)$ by reducing the OPS problem to the sequential probability assignment (SPA) problem [Cesa-Bianchi and Lugosi, 2006, Chapter 9.1]. Specifically, we consider the Kelly market vector setting, where $\mathbf{x}_t\in\{\mathbf{e}_1,\mathbf{e}_2\}$ for all $t\in[T]$, with $\mathbf{e}_1=(1,0)$ and $\mathbf{e}_2=(0,1)$. We encode the market outcomes by defining $y_t=1$ when $\mathbf{x}_t=\mathbf{e}_1$ and $y_t=0$ when $\mathbf{x}_t=\mathbf{e}_2$, and introduce the log loss

$$\ell_{\log}(w,y) = -y\log w - (1-y)\log(1-w).$$

It is straightforward to verify that $\ell_t(w)=\ell_{\log}(w,1)$ when $\mathbf{x}_t=\mathbf{e}_1$, and $\ell_t(w)=\ell_{\log}(w,0)$ when $\mathbf{x}_t=\mathbf{e}_2$. Consequently, let $\mathcal{Y}=\{0,1\}$, we obtain:

$$\mathcal{W}_T(\mathcal{U}_C) \ge \mathcal{V}_T(\mathcal{U}_C) := \inf_{w_1,\dots,w_T\in[0,1]} \sup_{y_1,\dots,y_T\in\mathcal{Y}} \sup_{u_{1:T}\in\mathcal{U}_C} \left(\sum_{t=1}^T \ell_{\log}(w_t,y_t) - \sum_{t=1}^T \ell_{\log}(u_t,y_t)\right).$$

**Hard Example Construction.** To lower bound the minimax regret $\mathcal{V}_T(\mathcal{U}_C)$, we construct an environment in which the time horizon is partitioned into consecutive intervals of length $L=\lceil 1/\varepsilon\rceil$ with $\varepsilon=\sqrt{C/T}$. Let $K=\lfloor T/L\rfloor$ denote the number of intervals. The first $K-1$ intervals each have length $L$, while the final interval has length $T-(K-1)L\in[L,2L-1]$.

For each block $k$, we consider two possible environments indexed by $I_k\in\{0,1\}$. The index $I_k$ is drawn independently and uniformly at random, i.e., $I_k\sim\mathrm{Bern}(1/2)$. For any $t\in\mathcal{I}_k$, we set $\widetilde{u}_t=\varepsilon I_k$, where $\varepsilon=\sqrt{C/T}$, so that $\widetilde{u}_t$ is fully determined by the environment index. The label is then generated as $y_t\sim\mathrm{Bern}(\widetilde{u}_t)$. For any realization of $\{I_K\}_{k=1}^K$, we have

$$\sum_{t=2}^T |\widetilde{u}_t - \widetilde{u}_{t-1}| \le \sum_{k=2}^K \varepsilon|I_k - I_{k-1}| \le \varepsilon(K-1)\le C,$$

which indicates $\widetilde{u}_{1:T}\in\mathcal{U}_C$. Then, the minimax regret $\mathcal{V}_T(\mathcal{U}_C)$ can be further lower bounded by

$$\begin{aligned}\mathcal{V}_T(\mathcal{U}_C) &\ge \inf_{w_{1:T}\in[0,1]^T} \mathbb{E}_{I_{1:K}}\left[\mathbb{E}_{y_{1:T}}\left[\sum_{t=1}^T \ell_{\log}(w_t,y_t) - \sum_{t=1}^T \ell_{\log}(\widetilde{u}_t,y_t)\,\middle|\, I_{1:K}\right]\right]\\ &= \inf_{w_{1:T}\in[0,1]^T} \mathbb{E}_{I_{1:K}}\left[\mathbb{E}_{y_{1:T}}\left[\log\left(\frac{\widetilde{\mathbf{q}}(y_{1:T})}{\mathbf{p}(y_{1:T})}\right)\,\middle|\, I_{1:K}\right]\right].\end{aligned}$$

In the above, we define $\mathbf{p}(y_{1:T})=\prod_{t=1}^T w_t^{y_t}\left(1-w_t\right)^{1-y_t}$. We note that the prediction $w_t$ is deterministic given $y_1,\dots,y_{t-1}$ and then the density function $\mathbf{p}$ is deterministic independent of the randomness in $I_{1:K}$ and $y_{1:T}$. Similarly, we define $\widetilde{\mathbf{q}}(y_{1:T})=\prod_{t=1}^T \widetilde{u}_t^{y_t}(1-\widetilde{u}_t)^{1-y_t}$. One can check that $\mathbf{p}(y_{1:T})$ and $\mathbf{q}(y_{1:T})$ are both valid probability density functions since

$$\sum_{y_{1:T}\in\{0,1\}^T} \mathbf{p}(y_{1:T}) = \sum_{y_{1:T-1}\in\{0,1\}^{T-1}} \mathbf{p}(y_{1:T-1}) = \cdots = \sum_{y_1\in\{0,1\}} \mathbf{p}(y_1) = 1.$$

Furthermore, since $y_t \sim \text{Bern}(\widetilde{u}_t)$ given $I_1, \ldots, I_K$, the joint distribution of the sequence $y_{1:T}$ has the same density function as $\widetilde{\mathbf{q}}$. Then, we have $\mathbb{E}_{y_{1:T}}\left[\log\left(\frac{\widetilde{\mathbf{q}}(y_{1:T})}{\mathbf{p}(y_{1:T})}\right) \mid I_{1:K}\right] = \text{KL}(\widetilde{\mathbf{q}}\|\mathbf{p})$. Then, the minimax regret can be further lower bounded by

$$\begin{aligned}
\mathcal{V}_T(\mathcal{U}_C) &\geq \inf_{w_{1:T}\in[0,1]^T} \mathbb{E}_{I_{1:K}}\left[\text{KL}(\widetilde{\mathbf{q}}\|\mathbf{p})\right] \\
&= \mathbb{E}_{I_{1:K}}\left[\text{KL}(\widetilde{\mathbf{q}}\|\bar{\mathbf{q}})\right] + \inf_{w_{1:T}\in[0,1]^T} \mathbb{E}_{I_{1:K}}\left[\sum_{y_{1:T}\in\{0,1\}^T} \widetilde{\mathbf{q}}(y_{1:T}) \ln \frac{\bar{\mathbf{q}}(y_{1:T})}{\mathbf{p}(y_{1:T})}\right] \\
&= \mathbb{E}_{I_{1:K}}\left[\text{KL}(\widetilde{\mathbf{q}}\|\bar{\mathbf{q}})\right] + \inf_{w_{1:T}\in[0,1]^T} \sum_{y_{1:T}\in\{0,1\}^T} \mathbb{E}_{I_{1:K}}\left[\widetilde{\mathbf{q}}(y_{1:T})\right] \ln \frac{\bar{\mathbf{q}}(y_{1:T})}{\mathbf{p}(y_{1:T})} \\
&= \mathbb{E}_{I_{1:K}}\left[\text{KL}(\widetilde{\mathbf{q}}\|\bar{\mathbf{q}})\right] + \inf_{w_{1:T}\in[0,1]^T} \text{KL}(\bar{\mathbf{q}}\|\mathbf{p}) \geq \mathbb{E}_{I_{1:K}}[\text{KL}(\widetilde{\mathbf{q}}\|\bar{\mathbf{q}})].
\end{aligned}$$

where $\bar{\mathbf{q}}(y_{1:T}) = \mathbb{E}_{I_{1:K}}\left[\widetilde{\mathbf{q}}(y_{1:T})\right]$ is the averaged comparator sequence. The last second line holds because both density functions $\bar{\mathbf{q}}$ and $\mathbf{p}$ are independent of $I_{1:K}$. Let $y_{\mathcal{I}_k} = \{y_t\}_{t\in\mathcal{I}_k}$ be the sequence of label on interval $\mathcal{I}_k$ and write their joint density as $\widetilde{\mathbf{q}}_k(y_{\mathcal{I}_k}) = \prod_{t\in\mathcal{I}_k} \widetilde{u}_t^{y_t}(1-\widetilde{u}_t)^{1-y_t}$. Since the label sequence $\{y_{\mathcal{I}_k}\}_{k=1}^K$ and block index $\{I_k\}_{k=1}^K$ are independent of each other, $\bar{\mathbf{q}}(y_{1:T}) = \prod_{k=1}^K \bar{\mathbf{q}}_k(y_{\mathcal{I}_k})$, where $\bar{\mathbf{q}}_k(y_{\mathcal{I}_k}) = \mathbb{E}_{I_k}[\widetilde{\mathbf{q}}_k(y_{\mathcal{I}_k})]$. The minimax regret can be further bounded by

$$\mathcal{V}_T(\mathcal{U}_C) \geq \mathbb{E}_{I_{1:K}}\left[\sum_{k=1}^K \text{KL}(\widetilde{\mathbf{q}}_k\|\bar{\mathbf{q}}_k)\right] = \sum_{k=1}^K \mathbb{E}_{I_K}[\text{KL}(\widetilde{\mathbf{q}}_k\|\bar{\mathbf{q}}_k)],$$

where the inequality follows from the additivity of the KL divergence for independent blocks, and the equality holds because $\widetilde{\mathbf{q}}_k$ depends only on $I_k$.

We next analyze the KL divergence contributed by each block. We have

$$\begin{aligned}
\mathbb{E}_{I_k}\left[\text{KL}\left(\widetilde{\mathbf{q}}_k \,\|\, \bar{\mathbf{q}}_k\right)\right] &= \frac{1}{2}\text{KL}\left(\mathbf{q}_1 \,\|\, \bar{\mathbf{q}}\right) + \frac{1}{2}\text{KL}\left(\mathbf{q}_2 \,\|\, \bar{\mathbf{q}}\right) \\
&\geq \sum_{y_{\mathcal{I}_k}} \frac{1}{4}\|\mathbf{q}_1(y_{\mathcal{I}_k}) - \mathbf{q}_2(y_{\mathcal{I}_k})\|_1^2 \\
&\geq \frac{1}{4}(1-(1-\varepsilon)^{|\mathcal{I}_k|})^2 \geq \frac{1}{4}(1-e^{-1})^2.
\end{aligned}$$

where $\mathbf{q}_1(y_{\mathcal{I}_k}) = \prod_{t\in\mathcal{I}_k} \varepsilon^{y_t}(1-\varepsilon)^{1-y_t}$ is the joint distribution of $|\mathcal{I}_k|$ independent Bernoulli distributions, $\mathbf{q}_2(y_{\mathcal{I}_k}) = \mathbb{1}\{y_t = 0, \forall t \in \mathcal{I}_k\}$ and $\bar{\mathbf{q}} = \mathbf{q}_1/2 + \mathbf{q}_2/2$ denotes their mixture. The second line is due to Pinsker's inequality. The third line is by taking the mass on $y_{\mathcal{I}_k} = 0, \ldots, 0$. The last second inequality is because $(1-\varepsilon)^{|\mathcal{I}_k|} \leq e^{-\varepsilon|\mathcal{I}_k|} \leq e^{-1}$ due to $|\mathcal{I}_k| \geq 1/\varepsilon$ for all $k \in [K]$. Then, we can conclude that

$$\mathcal{V}_T(\mathcal{U}_C) \geq \frac{(1-e^{-1})^2 K}{4} \geq \frac{(1-e^{-1})^2}{4}\left(\frac{1}{2}\sqrt{CT} - 1\right).$$

On the other hand, since any fixed comparator is also contained by $\mathcal{U}_C$, the standard lower bound for sequential probability assignment [Cesa-Bianchi and Lugosi, 2006] shows that $\mathcal{V}_T(\mathcal{U}_C) = \Omega(\log T)$ which leads to the lower bound of $\Omega(\max\{\log T, \sqrt{CT}\})$. □

## B.2 Proof of Lemma 1

*Proof.* For any interval $\mathcal{I} = [s, e] \subseteq [T]$, let $\mathbf{w}_{\mathcal{I}} = \arg\min_{\mathbf{w}\in\Delta_d} \sum_{t\in\mathcal{I}} \ell_t(\mathbf{w})$ be the optimal fixed prediction on interval $\mathcal{I}$. Then, for any sequence $\{\mathbf{u}_t\}_{t\in\mathcal{I}}$, we have

$$\sum_{t\in\mathcal{I}} \ell_t(\mathbf{w}_{\mathcal{I}}) - \sum_{t\in\mathcal{I}} \ell_t(\mathbf{u}_t) = \sum_{t\in\mathcal{I}} \ln\left(\frac{\mathbf{u}_t^\top \mathbf{x}_t}{\mathbf{w}_{\mathcal{I}}^\top \mathbf{x}_t}\right) = \ln\left(\frac{\prod_{t\in\mathcal{I}} \mathbf{u}_t^\top \mathbf{x}_t}{\prod_{t\in\mathcal{I}} \mathbf{w}_{\mathcal{I}}^\top \mathbf{x}_t}\right). \tag{13}$$

Let $z_i^{\max} = \max_{t\in\mathcal{I}}\{u_{t,i}\}$, and define $\mathbf{z}_{\max} = [z_1^{\max}, \dots, z_d^{\max}]^\top$ together with its normalized version $\bar{\mathbf{u}} = \mathbf{z}_{\max}/\|\mathbf{z}_{\max}\|_1$. Since $\mathbf{x}_t \in \mathbb{R}_+^d$, we can further upper bound (13) by

$$\ln\left(\frac{\prod_{t\in\mathcal{I}} \mathbf{u}_t^\top \mathbf{x}_t}{\prod_{t\in\mathcal{I}} \mathbf{w}_{\mathcal{I}}^\top \mathbf{x}_t}\right) \le \ln\left(\frac{\prod_{t\in\mathcal{I}} \mathbf{z}_{\max}^\top \mathbf{x}_t}{\prod_{t\in\mathcal{I}} \mathbf{w}_{\mathcal{I}}^\top \mathbf{x}_t}\right) = \ln\left(\frac{\prod_{t\in\mathcal{I}} \bar{\mathbf{u}}^\top \mathbf{x}_t}{\prod_{t\in\mathcal{I}} \mathbf{w}_{\mathcal{I}}^\top \mathbf{x}_t}\right) + |\mathcal{I}| \ln \|\mathbf{z}_{\max}\|_1 \le |\mathcal{I}| \ln \|\mathbf{z}_{\max}\|_1, \quad (14)$$

where the last inequality holds because $\mathbf{w}_{\mathcal{I}}$ minimizes the cumulative loss over the interval $\mathcal{I}$. For each coordinate of $\mathbf{z}_{\max}$, we have $z_i^{\max} \le u_{s,i} + \sum_{t=s+1}^{e} [u_{t,i} - u_{t-1,i}]_+$, which implies

$$\|\mathbf{z}_{\max}\|_1 \le \sum_{i=1}^{d} u_{s,i} + \sum_{t=s+1}^{e} \sum_{i=1}^{d} [u_{t,i} - u_{t-1,i}]_+ \le 1 + P'_{\mathcal{I}}, \quad (15)$$

where $P'_{\mathcal{I}} = \sum_{t=s+1}^{e} \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_1$ is the path-length measured in the $L_1$ norm.

Combining (13), (14), and (15) with condition (2), we obtain that for any interval $\mathcal{I} \subseteq [T]$,

$$\sum_{t\in\mathcal{I}} \ell_t(\mathbf{w}_t) - \sum_{t\in\mathcal{I}} \ell_t(\mathbf{u}_t) \le B(T) + |\mathcal{I}| \ln(1 + P'_{\mathcal{I}}) \le B(T) + |\mathcal{I}| P'_{\mathcal{I}}. \quad (16)$$

Next, we convert the interval-based dynamic regret bound (16) into a bound over the entire horizon by partitioning $[T]$ into $M$ consecutive intervals $\{\mathcal{I}_m\}_{m=1}^M$ such that $|\mathcal{I}_m| P'_{\mathcal{I}_m} \le B(T)$ for each block. The dynamic regret over $[T]$ can then be bounded as

$$\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) = \sum_{m=1}^{M} \left( \sum_{t\in\mathcal{I}_m} \ell_t(\mathbf{w}_t) - \sum_{t\in\mathcal{I}_m} \ell_t(\mathbf{u}_t) \right) \le 2MB(T). \quad (17)$$

It remains to show how to construct a partition $\{\mathcal{I}_m\}_{m=1}^M$ that satisfies $|\mathcal{I}_m| P'_{\mathcal{I}_m} \le B(T)$ using the minimum possible number of intervals $M$. This can be achieved via a virtual bin-partition scheme. Let $\mathcal{I}_m = [s_m, e_m]$ denote the $m$-th interval, with start time $s_m$ and end time $e_m$. For each $\tau > s_m$, we iteratively check whether $(\tau - s_m) P'_{[s_m,\tau]} \ge B(T)$, where $P'_{[s_m,\tau]} = \sum_{t=s_m+1}^{\tau} \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_1$. If the condition is violated at certain iteration $\tau$, we start a new interval by $s_{m+1} = \tau$; otherwise, we proceed to the next round.

Without loss of generality, we assume $M > 1$, i.e., the interval is restarted at least once; otherwise, the dynamic regret can be directly upper bounded by $B(T)$. By construction, this procedure guarantees that $|\mathcal{I}_m| P'_{\mathcal{I}_m} \le B(T)$ and $(e_m + 1 - s_m) P'_{[s_m,e_m+1]} > B(T)$ for the first $M-1$ intervals. Taking square roots on both sides and summing over all $M-1$ intervals yields

$$\begin{aligned}(M-1)\sqrt{B(T)} &\le \sum_{m=1}^{M} \sqrt{(e_m + 1 - s_m) P'_{[s_m,e_m+1]}} \\ &\le \sqrt{\left(\sum_{m=1}^{M} (e_m + 1 - s_m)\right)\left(\sum_{m=1}^{M} P'_{[s_m,e_m+1]}\right)} \le \sqrt{TP'_T},\end{aligned}$$

which implies $M \le 1 + \sqrt{TP'_T / B(T)}$. Substituting this bound into (17) completes the proof. □

# C Omitted Proofs for Section 2.3

## C.1 Useful Lemma

This section presents proofs that are omitted in Section 2. We first present the proof for Lemma 2 and then introduce a lemma that provides the upper bound on $\ln P_t(\mathbf{w})$ for any $\mathbf{w}$.

*Proof of Lemma 2.* This lemma holds by an exact identity for distributions defined over $\Delta_d$. Let $Z_t = \mathbb{E}_{\mathbf{w}\sim P_t}[\exp(-\ell_t(\mathbf{w}))]$, we have

$$\begin{aligned}\mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_t \,\|\, P'_{t+1}\right) &= \mathbb{E}_{\mathbf{u}\sim Q_t}\left[\ln\left(\frac{P'_{t+1}(\mathbf{u})}{P_t(\mathbf{u})}\right)\right] = \mathbb{E}_{\mathbf{u}\sim Q_t}\left[\ln\left(\frac{\exp(-\ell_t(\mathbf{u}))}{Z}\right)\right] \\ &= -\mathbb{E}_{\mathbf{u}\sim Q_t}\left[\ell_t(\mathbf{u})\right] - \ln Z = -\mathbb{E}_{\mathbf{u}\sim Q_t}\left[\ell_t(\mathbf{u})\right] + \ell_t(\mathbf{w}_t),\end{aligned}$$

where the second inequality is because $P'_{t+1}(\mathbf{u}) = P_t(\mathbf{u}) \cdot \exp(-\ell_t(\mathbf{u}))/Z$. □

**Lemma 4.** *Let $P_t$ be the distribution updated in Algorithm 1 at round $t$. Then, for any $\mu \in [0,1]$, we have $\ln P_t(\mathbf{u}) \le (d-1)\ln(T+1) + d(\ln d + 1)$ for all $\mathbf{u} \in \Delta_d$.*

*Proof.* According to the EW and fixed-share update step in Algorithm 1, for any $\mathbf{u} \in \Delta_d$, we have

$$P_{t+1}(\mathbf{w}) = (1-\mu)\widetilde{P}_{t+1}(\mathbf{w}) + \mu P_1(\mathbf{w}) = \frac{(1-\mu)\cdot e^{-\ell_t(\mathbf{w})}}{\mathbb{E}_{\mathbf{u}\sim P_t}[e^{-\ell_t(\mathbf{u})}]} P_t(\mathbf{w}) + \mu P_1(\mathbf{w}).$$

Let $Z_t = \mathbb{E}_{\mathbf{u}\sim P_t}[e^{-\ell_t(\mathbf{u})}]$. By unrolling the recursive definition of $P_t$, we have

$$P_{t+1}(\mathbf{w}) = \sum_{i=2}^{t+1} \frac{\mu(1-\mu)^{t-i+1}}{\prod_{s=i}^{t} Z_s} \cdot P_1(\mathbf{w}) e^{-\sum_{s=i}^{t}\ell_s(\mathbf{w})} + \frac{(1-\mu)^t}{\prod_{s=1}^{t} Z_s} \cdot P_1(\mathbf{w}) e^{-\sum_{s=1}^{t}\ell_s(\mathbf{w})}, \tag{18}$$

where for the case $i = t+1$, we adopt the convention that the empty sum and empty product satisfy $\sum_{s=t+1}^{t}\ell_s(\mathbf{w}) = 0$ and $\prod_{s=t+1}^{t} Z_s = 1$. Let us further define the individual distribution

$$P_{t+1,i}(\mathbf{w}) = \frac{1}{W_{t+1,i}} P_1(\mathbf{w}) e^{-\sum_{s=i}^{t}\ell_s(\mathbf{w})} \text{ for all } i \le t+1, \tag{19}$$

where $W_{t+1,i} = \mathbb{E}_{\mathbf{u}\sim P_1}[e^{-\sum_{s=i}^{t}\ell_s(\mathbf{u})}]$ is the normalization factor. Then, we can further rewrite $P_{t+1}(\mathbf{w})$ in (18) as a weighted combination of the distributions $P_{t+1,i}(\mathbf{w})$:

$$P_{t+1}(\mathbf{w}) = \sum_{i=1}^{t+1} \alpha_{t+1,i} P_{t+1,i}(\mathbf{w}) \le \max_{i\in[t+1]} P_{t+1,i}(\mathbf{w}). \tag{20}$$

In the above, the weight can be calculated by $\alpha_{t+1,i} = \mu(1-\mu)^{t-i+1} W_{t+1,i} / \prod_{s=i}^{t} Z_s$ for $i \in [2, t+1]$ and $\alpha_{t+1,1} = (1-\mu)^t W_{t+1,1} / \prod_{s=1}^{t} Z_s$. Since both $P_{t+1}$ and $P_{t+1,i}$ are probability distributions, we have $\sum_{i=1}^{t+1}\alpha_{t+1,i} = 1$. The last inequality in (20) holds due to Hölder's inequality.

For $i = t+1$, we have $P_{t+1,t+1}(\mathbf{w}) = P_1(\mathbf{w})$ for all $\mathbf{w} \in \Delta_d$ and $\ln P_{t+1,t+1}(\mathbf{w}) = \ln\Gamma(d)$ since $P_1$ is a uniform distribution on $\Delta_d$. Then, to prove the lemma it is sufficient to upper bound $\ln P_{t+1,i}(\mathbf{w})$ for all $i \in [t]$ and $\mathbf{w} \in \Delta_d$. According to the definition (19), the distribution $P_{t+1,i}(\mathbf{w}) \propto P_{t,i}(\mathbf{w}) \cdot \exp(-\ell_t(\mathbf{w}))$ for all $i \le t$. Then, we have

$$\begin{aligned}
\ln P_{t+1,i}(\mathbf{w}) &= \ln P_{t,i}(\mathbf{w}) - \ln(\mathbb{E}_{\mathbf{u}\sim P_{t,i}}[e^{-\ell_t(\mathbf{u})}]) - \ell_t(\mathbf{w}) \\
&= \ln P_1(\mathbf{w}) - \sum_{s=i}^{t} \ln(\mathbb{E}_{\mathbf{u}\sim P_{s,i}}[e^{-\ell_s(\mathbf{u})}]) - \sum_{s=i}^{t}\ell_s(\mathbf{w}) \\
&= \ln P_1(\mathbf{w}) + \sum_{s=i}^{t}\mathbb{E}_{\mathbf{w}\sim Q}[\ell_s(\mathbf{u})] - \sum_{s=i}^{t}\ell_s(\mathbf{w}) + \mathrm{KL}\left(Q \,\|\, P_1\right) - \mathrm{KL}\left(Q \,\|\, P_{t+1,i}\right)
\end{aligned} \tag{21}$$

for any $Q \in \Delta_d$, where the second equality is due to the recursive definition of $P_{t,i}$. Let $\mathbf{w}_*^{t+1,i} = \arg\max_{\mathbf{w}\in\Delta_d} \ln P_{t+1,i}(\mathbf{w})$. We can choose $Q = Q_*^{t+1,i} = \mathrm{Dir}(\mathbf{1} + T\mathbf{w}_*^{t+1,i})$. Then, the same arguments in upper bounding term (a) in the proof of Theorem 2 yields

$$\sum_{s=i}^{t}\mathbb{E}_{\mathbf{w}\sim Q_*^{t+1,i}}[\ell_s(\mathbf{u})] - \sum_{s=i}^{t}\ell_s(\mathbf{w}_*^{t+1,i}) \le T\ln\left(1 + \frac{d}{T}\right) \le d.$$

Besides, based on Lemma 3, we have $\mathrm{KL}(Q_*^{t+1,i} \,\|\, P_1) \le (d-1)\ln(1+T)$. Plugging the above inequalities into (21) with the fact that $\ln P_1(\mathbf{w}) = \ln\Gamma(d)$ yields

$$\ln P_{t+1,i}(\mathbf{w}) \le \ln\Gamma(d) + d + (d-1)\ln(1+T) = (d-1)\ln(1+T) + d(\ln d + 1)$$

for any $\mathbf{w} \in \Delta_d$ and $i \in [t]$. We end the proof by the fact $P_{t+1}(\mathbf{w}) \le \max_{i\in[t+1]} P_{t+1,i}(\mathbf{w})$. □

## C.2 Proof of Theorem 2

*Proof.* Let $P_1 = \mathrm{Dir}(\mathbf{1})$. According to the definition of the loss function $\ell_t(\mathbf{w}) = -\ln(\mathbf{w}^\top \mathbf{x}_t)$ and the prediction $\mathbf{w}_t = \mathbb{E}_{\mathbf{w}\sim P_t}[\mathbf{w}]$, we have

$$\begin{aligned}
\ell_t(\mathbf{w}_t) &= -\ln\left(\mathbb{E}_{\mathbf{w}\sim P_t}[\exp(-\ell_t(\mathbf{w}))]\right) \\
&= \mathbb{E}_{\mathbf{w}\sim Q_t}[\ell_t(\mathbf{w})] + \mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_t \,\|\, \widetilde{P}_{t+1}\right) \\
&= \mathbb{E}_{\mathbf{w}\sim Q_t}[\ell_t(\mathbf{w})] + \mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_t \,\|\, P_{t+1}\right) + \mathbb{E}_{\mathbf{w}\sim Q_t}\left[\ln \frac{\widetilde{P}_{t+1}(\mathbf{w})}{P_{t+1}(\mathbf{w})}\right] \\
&\le \mathbb{E}_{\mathbf{w}\sim Q_t}[\ell_t(\mathbf{w})] + \mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_t \,\|\, P_{t+1}\right) + \ln \frac{1}{1-\mu_{t+1}}
\end{aligned}$$

where $\widetilde{P}_{t+1}(\mathbf{w}) \propto P_t(\mathbf{w})\exp(-\ell_t(\mathbf{w}))$, $\forall \mathbf{u} \in \Delta_d$. In the above, the last inequality holds since $P_{t+1}(\mathbf{w}) = (1-\mu_{t+1})\widetilde{P}_{t+1}(\mathbf{w}) + \mu_{t+1}P_1(\mathbf{w})$ for all $\mathbf{w} \in \Delta_d$ and $t \ge 1$. We can further upper bound the term $\ln(1/(1-\mu_{t+1})) \le 1/t$ by the setting $\mu_{t+1} = 1/(1+t)$. Taking the sum from $t = 1$ to $T$ rounds and rearranging the terms yields

$$\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) \le \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathbf{w}\sim Q_t}[\ell_t(\mathbf{w})]}_{\texttt{term (a)}} + \underbrace{\sum_{t=2}^{T} \left(\mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_{t-1} \,\|\, P_t\right)\right)}_{\texttt{term (b)}} + \underbrace{\mathrm{KL}\left(Q_1 \,\|\, P_1\right)}_{\texttt{term (c)}} + (1 + \ln T).$$

The next step is to choose $Q_t$ to make the bound tight. Here, we use the Dirichlet distribution $Q_t = \mathrm{Dir}(\mathbf{1} + \gamma \mathbf{u}_t)$, where $\gamma > 0$ is a parameter to be tuned later and $\mathbf{u}_t$ is the time-varying comparator sequence. Next, we bound the three terms separately.

**Bounding term (a).** By Assumption 1, for each round $t \in [T]$, the return vector $\mathbf{x}_t$ has at least one nonzero entry. For simplicity, we assume that the first $\bar{d}_t$ entries of $\mathbf{x}_t$ are nonzero. Besides, for each dimension $i \in [d]$, let $\{Z_{t,i}\}_{i=1}^d$ be independent random variables with $Z_{t,i} \sim \mathrm{Gamma}(1 + \gamma u_{t,i}, 1)$, and define $Z_{t,0} = \sum_{i=1}^d Z_{t,i} \sim \mathrm{Gamma}(d+\gamma, 1)$. It is known that the random vector $(Z_{t,1}/Z_{t,0}, \ldots, Z_{t,d}/Z_{t,0})$ follows the Dirichlet distribution $\mathrm{Dir}(\mathbf{1} + \gamma \mathbf{u}_t)$. We have

$$\begin{aligned}
\texttt{term (a)} &= \sum_{t=1}^{T} \mathbb{E}_{Z_{t,1},\ldots,Z_{t,d}}\left[-\ln\left(\sum_{i=1}^{d} \frac{x_{t,i} Z_{t,i}}{Z_{t,0}}\right)\right] \\
&= \sum_{t=1}^{T} \mathbb{E}_{Z_{t,0}}[\ln(Z_{t,0})] + \sum_{t=1}^{T} \mathbb{E}_{Z_{t,1},\ldots,Z_{t,d}}\left[-\ln\left(\sum_{i=1}^{d} x_{t,i} Z_{t,i}\right)\right] \\
&= \sum_{t=1}^{T} \psi(d+\gamma) + \sum_{t=1}^{T} \mathbb{E}_{Z_{t,1},\ldots,Z_{t,d}}\left[-\ln\left(\sum_{i=1}^{\bar{d}_t} x_{t,i} Z_{t,i}\right)\right] \\
&\le \sum_{t=1}^{T} \psi(d+\gamma) + \sum_{t=1}^{T} \mathbb{E}_{Z_{t,1},\ldots,Z_{t,d}}\left[-\sum_{i=1}^{\bar{d}_t} p_{t,i} \ln(x_{t,i} Z_{t,i}/p_{t,i})\right] \\
&= \sum_{t=1}^{T} \psi(d+\gamma) - \sum_{t=1}^{T}\left(\sum_{i=1}^{\bar{d}_t} p_{t,i}\big(\ln x_{t,i} + \psi(1+\gamma u_{t,i})\big) - \sum_{i=1}^{\bar{d}_t} p_{t,i} \ln(p_{t,i})\right).
\end{aligned} \tag{22}$$

In the third line, we use the identity $\mathbb{E}[\ln(Z_{t,0})] = \psi(d+\gamma)$ for $Z_{t,0} \sim \mathrm{Gamma}(d+\gamma, 1)$, and note that $\sum_{i=1}^d x_{t,i} Z_{t,i} = \sum_{i=1}^{\bar{d}} x_{t,i} Z_{t,i}$ since the remaining entries are zero. For the fourth line, since $x_{t,i} Z_{t,i} > 0$ for all $i \in [\bar{d}_t]$ and $t \in [T]$, we apply the inequality $\ln\left(\sum_{i=1}^{\bar{d}_t} a_i\right) \ge \sum_{i=1}^{\bar{d}_t} p_i \ln\left(\frac{a_i}{p_i}\right)$, which holds for all $a_i > 0$ and any probability vector $\mathbf{p} = (p_1, \ldots, p_{\bar{d}_t}) \in \mathrm{ri}(\Delta_{\bar{d}_t})$, where $\mathrm{ri}(\Delta_{\bar{d}_t})$ denotes the relative interior of the simplex $\Delta_{\bar{d}_t}$.

Then, we can tune the probability vector $\mathbf{p}_t$ to make the bound (22) tight. The goal is to solve the optimization problem

$$V_t^* = \max_{\mathbf{p}_t \in \mathrm{ri}(\Delta_{\bar{d}_t})} \sum_{i=1}^{\bar{d}_t} p_{t,i}\big(\ln x_{t,i} + \psi(1+\gamma u_{t,i})\big) - \sum_{i=1}^{\bar{d}_t} p_{t,i} \ln(p_{t,i}),$$

which has the closed-form solution by $V_t^* = \ln\left(\sum_{i=1}^{\bar{d}_t} x_{t,i} \exp(\psi(1+\gamma u_{t,i}))\right)$ achieved at $p_{t,i}^* \propto x_{t,i}\exp(\psi(1+\gamma u_{t,i}))$. Plugging the optimal solution back to (22) yields

$$\begin{aligned}
\texttt{term (a)} &\le \sum_{t=1}^{T} \psi(d+\gamma) - \sum_{t=1}^{T} \ln\left(\sum_{i=1}^{\bar{d}_t} x_{t,i}\exp(\psi(1+\gamma u_{t,i}))\right) \\
&\le \sum_{t=1}^{T} \psi(d+\gamma) - \sum_{t=1}^{T} \ln\left(\gamma \sum_{i=1}^{\bar{d}_t} x_{t,i} u_{t,i}\right) \\
&= \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) + T(\psi(d+\gamma) - \ln\gamma) \\
&\le \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) + T\ln\left(1+\frac{d}{\gamma}\right)
\end{aligned}$$

where the second line holds since $\psi(1+x) \ge \ln x$ for all $x > 0$. The third line follows from the fact that only the first $\bar{d}$ entries of $\mathbf{x}_t$ are nonzero. The last inequality holds since $\psi(x) \le \ln(x)$ for all $x > 0$.

**Bounding term (b):** As for term (b), we have

$$\begin{aligned}
\texttt{term (b)} &= \sum_{t=2}^{T} \left(\mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_{t-1} \,\|\, P_t\right)\right) \\
&= \sum_{t=2}^{T} \mathbb{E}_{\mathbf{w}\sim Q_t}\left[\ln\frac{Q_t(\mathbf{w})}{P_t(\mathbf{w})}\right] - \mathbb{E}_{\mathbf{w}\sim Q_{t-1}}\left[\ln\frac{Q_{t-1}(\mathbf{w})}{P_t(\mathbf{w})}\right] \\
&= \sum_{t=2}^{T} \left(H(Q_{t-1}) - H(Q_t)\right) + \sum_{t=2}^{T} \mathbb{E}_{\mathbf{w}\sim Q_{t-1}}\left[\ln P_t(\mathbf{w})\right] - \mathbb{E}_{\mathbf{w}\sim Q_t}\left[\ln P_t(\mathbf{w})\right] \\
&= \underbrace{\sum_{t=2}^{T} \int_{\mathbf{w}\in\Delta_d} (Q_{t-1}(\mathbf{w}) - Q_t(\mathbf{w}))\ln P_t(\mathbf{w})\mathrm{d}\mathbf{w}}_{\texttt{term (b-1)}} + \underbrace{H(Q_1) - H(Q_T)}_{\texttt{term (b-2)}}
\end{aligned}$$

We then proceed to bound term (b-1) and term (b-2) separately. For notational simplicity, we denote by $C_{T,d} = (d-1)\ln(T+1) + d(\ln d + 1)$. We can upper bound term (b-1) by

$$\begin{aligned}
\texttt{term (b-1)} &\le \sum_{t=2}^{T} \int_{\mathbf{w}\in\Delta_d} |Q_{t-1}(\mathbf{w}) - Q_t(\mathbf{w})| \cdot |\ln P_t(\mathbf{w})| \mathrm{d}\mathbf{w} \\
&\le C_{T,d} \sum_{t=2}^{T} \int_{\Delta_d} |Q_{t-1}(\mathbf{w}) - Q_t(\mathbf{w})| \mathrm{d}\mathbf{w}, \qquad (23)
\end{aligned}$$

where the last inequality holds because $\ln P_t(\mathbf{w}) \le C_{T,d}$ for all $\mathbf{w} \in \Delta_d$ by Lemma 4, and $\ln(1/P_t(\mathbf{w})) \le \ln(1/(\mu_t P_1(\mathbf{w}))) = \ln(T/\Gamma(d-1))$ for all $\mathbf{w} \in \Delta_d$ due to the fixed-share update (4). Therefore, we obtain $|\ln P_t(\mathbf{w})| \le \max\{C_{T,d}, \ln(T/\Gamma(d-1))\} \le C_{T,d}$. Then, we can further bounded the total variation

between $Q_t$ and $Q_{t-1}$ by

$$
\begin{aligned}
&\sum_{t=2}^{T} \int_{\Delta_d} |Q_{t-1}(\mathbf{w}) - Q_t(\mathbf{w})| \mathrm{d}\mathbf{w} \\
&\leq \sum_{t=2}^{T} \sqrt{\frac{1}{4}(\mathrm{KL}(Q_t \| Q_{t-1}) + \mathrm{KL}(Q_{t-1} \| Q_t))} \\
&= \sum_{t=2}^{T} \sqrt{\gamma \sum_{i=1}^{d} (u_{t,i} - u_{t-1,i}) \big(\psi(1+\gamma u_{t,i}) - \psi(1+\gamma u_{t-1,i})\big)} \\
&= \sum_{t=2}^{T} \sqrt{\gamma^2 \sum_{i=1}^{d} (u_{t,i} - u_{t-1,i})^2 \psi'(1+\gamma \xi_{t,i})} \\
&\leq \sum_{t=2}^{T} \sqrt{\sum_{i=1}^{d} \frac{2\gamma^2}{1+\gamma\xi_{t,i}} (u_{t,i} - u_{t-1,i})^2} \leq \sqrt{2} \sum_{t=2}^{T} \min\left\{\sqrt{\frac{\gamma}{\alpha}}, \gamma\right\} \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2
\end{aligned} \tag{24}
$$

where the first inequality holds by Pinsker's inequality. The second line holds by the closed formulation of KL divergence between two Dirichlet distributions in Proposition 2. The second equality is due to the mean value theorem, where $\xi_{t,i}$ is a point on the line connecting $u_{t,i}$ and $u_{t-1,i}$. The fourth line holds because the trigamma function $\psi'(\alpha) \leq \frac{2}{\alpha}$ for $\alpha \geq 1$. The last inequality is due to $\xi_{t,i} \geq \min\{u_{t,i}, u_{t-1,i}\} \geq \alpha$ such that $\frac{\gamma^2}{1+\gamma\xi_{t,i}} \leq \min\{\gamma^2, \frac{\gamma}{\alpha}\}$. Combining (23) and (24), we arrive

$$
\texttt{term (b-1)} \leq \sqrt{2} C_{T,d} \min\left\{\sqrt{\frac{\gamma}{\alpha}}, \gamma\right\} P_T,
$$

where $P_T = \sum_{t=2}^{T} \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$. As for term (b-2), we have

$$
\texttt{term (b-2)} = H(Q_1) - H(Q_T) = \mathrm{KL}\left(Q_T \,\|\, P_1\right) - \mathrm{KL}\left(Q_1 \,\|\, P_1\right)
$$

since $P_1 = \mathrm{Dir}(\mathbf{1})$ is a uniform distribution over the simplex. Finally, we arrive

$$
\texttt{term (b)} \leq \sqrt{2} C_{T,d} \min\left\{\sqrt{\frac{\gamma}{\alpha}}, \gamma\right\} P_T + \mathrm{KL}\left(Q_T \,\|\, P_1\right) - \mathrm{KL}\left(Q_1 \,\|\, P_1\right)
$$

**Combining All**. Combining the bounds on terms (a) and (b), we get

$$
\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) \leq \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) + T \ln\left(1 + \frac{d}{\gamma}\right) + \sqrt{2} C_{T,d} \min\left\{\sqrt{\frac{\gamma}{\alpha}}, \gamma\right\} P_T + \mathrm{KL}\left(Q_T \,\|\, P_1\right) + 2. \tag{25}
$$

We can further upper bound the KL divergence term by

$$
\begin{aligned}
\mathrm{KL}\left(Q_T \,\|\, P_1\right) \leq \mathrm{KL}\left(\mathbf{1} + \gamma \mathbf{e}_i \,\|\, P_1\right) &= \ln \frac{\Gamma(d+\gamma)}{\Gamma(d)\Gamma(1+\gamma)} + \gamma(\psi(1+\gamma) - \psi(d+\gamma)) \\
&\leq (d-1)\ln(1+\gamma).
\end{aligned}
$$

where $\mathbf{e}_i$ is a one-hot vector with 1 at the $i$-th position and 0 elsewhere. The first inequality is due to Lemma 3. The last inequality holds by $\Gamma(d+\gamma)/\Gamma(1+\gamma) = \prod_{i=1}^{d-1}(r+i)$ and $\gamma > 0$. Then, the regret bound (25) becomes

$$
\begin{aligned}
&\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) \\
&\leq T \ln\left(1 + \frac{d}{\gamma}\right) + \sqrt{2} C_{T,d} \min\left\{\sqrt{\frac{\gamma}{\alpha}}, \gamma\right\} P_T + (d-1)\ln(1+\gamma) + (1 + \ln T),
\end{aligned}
$$

where $C_{T,d} = (d-1)\ln(T+1) + d(\ln d + 1) = \mathcal{O}\big(d \ln(Td)\big)$. Since $\gamma$ is a free parameter of the virtual comparator sequence $Q_t$, we can tune $\gamma$ to tighten the regret bound. We consider the following two cases, depending on the value of $P_T$:

- **Case 1 ($P_T \le 1/T$).** We set $\gamma = T$, which yields an $\mathcal{O}(d\ln(dT))$ regret bound.
- **Case 2 ($P_T > 1/T$).** We choose $\gamma = \alpha^{\frac{1}{3}} T^{\frac{2}{3}} P_T^{-\frac{2}{3}} (\ln(dT))^{-\frac{2}{3}}$. Substituting this choice of $\gamma$ into the bound gives

$$\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) = \mathcal{O}\left(d\Big(\alpha^{-\frac{1}{3}} T^{\frac{1}{3}} P_T^{\frac{2}{3}} (\ln(dT))^{\frac{2}{3}} + \ln T + \ln(1/\alpha)\Big)\right).$$

We have completed the proof by combining the two cases. □

## C.3 Proof of Corollary 1

*Proof.* Fix any comparator sequence $\mathbf{u}_1, \ldots, \mathbf{u}_T \in \Delta_d$ and let $\beta \in (0,1)$ be a certain parameter for mixing the comparator. We define the interior counterpart of $\mathbf{u}_t$ by $\widetilde{\mathbf{u}}_t := (1-\beta)\mathbf{u}_t + \frac{\beta}{d}\mathbf{1}$ for all $t \in [T]$. Clearly, we have $\min_{i\in[d]} \widetilde{u}_{t,i} \ge \beta/d$. Besides, the gap between the $\widetilde{\mathbf{u}}_t$ and $\mathbf{u}_t$ can be bounded by

$$\begin{aligned}\sum_{t=1}^{T} \ell_t(\widetilde{\mathbf{u}}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) &= \sum_{t=1}^{T} \ln\left(\frac{\mathbf{u}_t^\top \mathbf{x}_t}{(1-\beta)\mathbf{u}_t^\top \mathbf{x}_t + \frac{\beta}{d}\mathbf{1}^\top \mathbf{x}_t}\right) \\ &\le \sum_{t=1}^{T} \ln\left(\frac{\mathbf{u}_t^\top \mathbf{x}_t}{(1-\beta)\mathbf{u}_t^\top \mathbf{x}_t}\right) = T\ln\left(\frac{1}{1-\beta}\right).\end{aligned}$$

Then, we can upper bound the dynamic regret of the algorithm with respect to any comparator $\mathbf{u}_t \in \Delta$ by

$$\begin{aligned}\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) &\le \sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\widetilde{\mathbf{u}}_t) + \sum_{t=1}^{T} \ell_t(\widetilde{\mathbf{u}}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) \\ &\le \sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\widetilde{\mathbf{u}}_t) \;+\; T\ln\left(\frac{1}{1-\beta}\right) \\ &\le \beta^{-\frac{1}{3}}(1-\beta)^{\frac{2}{3}} d^{\frac{4}{3}} B T^{\frac{1}{3}} P_T^{\frac{2}{3}} + dB\ln\left(\frac{dT}{\beta}\right) + T\ln\left(\frac{1}{1-\beta}\right),\end{aligned}$$

where the last line is by the dynamic regret guarantee for the sequence in the relative interior. Since the guarantee of the algorithm $\mathcal{A}$ holds for any $\beta \in (0,1)$, we can choose $\beta$ to make the bound tight. For simplicity, we assume $T > 2d$, then the $\beta$ can be tuned separately in the following three cases:

- **Case 1 ($P_T \le 1/T$).** Set $\beta = d/T$, then we have $\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}(dB\ln(dT) + d)$.
- **Case 2 $\big(1/T < P_T \le T/(4d^2B^{3/2})\big)$.** We set $\beta = dB^{\frac{3}{4}}\sqrt{\frac{P_T}{T}}$, which ensures $\beta \le 1/2$ under the condition of this case. Using $(1-\beta)^{2/3} \le 1$ and $T\ln(\frac{1}{1-\beta}) \le 2\beta T$, we obtain

$$\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) = \mathcal{O}\left(dB^{\frac{3}{4}}\sqrt{TP_T} + dB\ln(dT)\right).$$

- **Case 3 $\big(P_T > T/(4d^2B^{3/2})\big)$.** We set $\beta = 1 - d^{-\frac{1}{2}}$. In such a case, we have $1-\beta = d^{-1/2}$ and $\beta \ge 1/2$. The dynamic regret can be upper bounded by

$$\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) = O\left(d\,B\,T^{\frac{1}{3}} P_T^{\frac{2}{3}} + dB^{\frac{3}{4}}\sqrt{TP_T}\ln d + dB\ln(dT)\right),$$

  where we use $T \le 2dB^{3/4}\sqrt{TP_T}$ by the condition of $P_T$ in this case.

We have completed the proof. □

### C.4 Proof of Proposition 1

*Proof of Proposition 1.* Following the same steps as in the proof of Theorem 2, for any distribution $Q_t \in \Delta_d$ we have

$$\ell_t(\mathbf{w}_t) \le \mathbb{E}_{\mathbf{w}\sim Q_t}[\ell_t(\mathbf{w})] + \mathrm{KL}\left(Q_t \,\|\, P_t\right) - \mathrm{KL}\left(Q_t \,\|\, P_{t+1}\right) + \ln \frac{1}{1-\mu_{t+1}}.$$

For each round $t \in \mathcal{I} = [r, s]$, we take $Q_t = Q_{\mathcal{I}} = \mathrm{Dir}(\mathbf{1} + \gamma \mathbf{u})$, i.e., a fixed Dirichlet distribution independent of $t$. Summing the above inequality over the interval $\mathcal{I} = [r, s]$ with $\mu_t = 1/t$ yields

$$\sum_{t\in\mathcal{I}} \ell_t(\mathbf{w}_t) \le \underbrace{\sum_{t\in\mathcal{I}} \mathbb{E}_{\mathbf{w}\sim Q_{\mathcal{I}}}[\ell_t(\mathbf{w})]}_{\texttt{term (a)}} + \underbrace{\mathrm{KL}\left(Q_{\mathcal{I}} \,\|\, P_r\right)}_{\texttt{term (b)}} - \mathrm{KL}\left(Q_{\mathcal{I}} \,\|\, P_{s+1}\right) + \sum_{t\in\mathcal{I}} \ln\left(1 + \frac{1}{t}\right).$$

For term (a), following the same steps as in the proof of Theorem 2, we have

$$\texttt{term (a)} \le \sum_{t\in\mathcal{I}} \ell_t(\mathbf{u}) + |\mathcal{I}| \ln\left(1 + \frac{d}{\gamma}\right).$$

As for term (b), we have

$$\begin{aligned}
\texttt{term (b)} &= \mathrm{KL}\left(Q_{\mathcal{I}} \,\|\, P_r\right) \\
&= \mathrm{KL}\left(Q_{\mathcal{I}} \,\|\, P_1\right) + \mathbb{E}_{\mathbf{w}\sim Q_{\mathcal{I}}}\left[\ln \frac{P_1(\mathbf{w})}{P_r(\mathbf{w})}\right] \\
&\le \mathrm{KL}\left(Q_{\mathcal{I}} \,\|\, P_1\right) + \ln \frac{1}{\mu_r} \le (d-1)\ln(1+\gamma) + \ln T.
\end{aligned}$$

where the first inequality holds since $P_r(\mathbf{w}) \ge \mu_r P_1(\mathbf{w})$ by the fixed share update. We complete the proof by combining the upper bound of terms (a) and (b) with the setting $\gamma = T$. □

## D Omitted Proofs for Section 3

*Proof of Theorem 3.* We begin with a similar mixability-based regret decomposition as Zhang et al. [2025]. Let $\widetilde{m}_t(P_t) = -\frac{1}{\eta} \ln\left(\mathbb{E}_{\mathbf{u}\sim P_t}[e^{-\eta \widetilde{\ell}_t(\mathbf{u})}]\right)$ be the mix loss. The dynamic regret can be decomposed by

$$\begin{aligned}
&\sum_{t=1}^{T} \ell_t(\mathbf{w}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{u}_t) \le \sum_{t=1}^{T} \widetilde{\ell}_t(\mathbf{w}_t) - \sum_{t=1}^{T} \widetilde{\ell}_t(\mathbf{u}_t) \\
&= \underbrace{\sum_{t=1}^{T} \widetilde{\ell}_t(\mathbf{w}_t) - \sum_{t=1}^{T} \widetilde{m}_t(P_t)}_{\texttt{term (a)}} + \underbrace{\sum_{t=1}^{T} \widetilde{m}_t(P_t) - \sum_{t=1}^{T} \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})]}_{\texttt{term (b)}} + \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] - \sum_{t=1}^{T} \widetilde{\ell}_t(\mathbf{u})}_{\texttt{term (c)}},
\end{aligned}$$

where the first line is due to Hazan [2016, Lemma 4.7] under the step size setting $\eta = \frac{1}{5} \min\left\{\frac{1}{2G}, 1\right\}$. In the analysis we choose $Q_t = \mathcal{N}(\mathbf{u}_t, \sigma^2 I_d)$, where $\sigma \in (0, 1]$ is a parameter that can be virtually tuned to make the bound tight.

One can handle terms (a) and (c) using arguments similar to those in Zhang et al. [2025]. However, the most challenging part is the analysis of term (b), where the domain constraint is enforced via an intractable I-projection of the distribution $P_t$ onto a set of infinite Gaussian mixtures. In our algorithm, instead of performing this intractable projection, we identify that it is sufficient to project each component of the mixture distribution $P_t$ individually rather than projecting the mixture as a whole. The latter would require a more in-depth analysis that leverages the two-layer structure of Algorithm 2. This leads to an efficient method. In what follows, we first analyze terms (a) and (c) using arguments similar to those in [Zhang et al., 2025], and then turn to the most challenging term (b).

**Bounding term (a).** Since $\widetilde{\ell}_t(\mathbf{w})$ is a quadratic function and the initial distribution of each base base-leaner $\mathcal{B}_i$ is a Gaussian, according to van der Hoeven et al. [2018, Theorem 5] shows that the distribution $P_{t+1,i} = \mathcal{N}(\mathbf{w}_{t+1,i}, H_{t+1,i}^{-1})$ for any base-learner $B_i$ updated by (5) is also a Gaussian distribution. More precisely, the mean and covariance matrix can be updated by

$$\begin{cases} H_{t+1,i} = H_{t,i} + 2\eta^2 \mathbf{g}_t \mathbf{g}_t^\top \\ \mathbf{w}'_{t+1,i} = \mathbf{w}_{t,i} - \eta H_{t+1,i}^{-1}(1 - 2\eta \mathbf{g}_t^\top (\mathbf{w}_t - \mathbf{w}_{t,i}))\mathbf{g}_t \\ \mathbf{w}_{t+1,i} = \arg\min_{\mathbf{u}\in\mathcal{W}} \|\mathbf{u} - \mathbf{w}'_{t+1,i}\|_{H_{t+1,i}} \end{cases} \tag{26}$$

The above essentially follows the update procedure of online Newton step [Hazan et al., 2007]. The design matrix is also symmetric positive definite and $H_{t+1,i} = I_d + 2\eta^2 \sum_{t=i}^{t} \mathbf{g}_t \mathbf{g}_t^\top \leq (1 + \frac{dt}{2}) I_d \leq dT I_d$ for any $\mathcal{B}_i \in H_t$ and $\mathbf{w}_{t,i} \in \mathcal{W}$ due to the projection step.

Our goal is to show `term (a)` $\leq 0$. To show this, it is sufficient to have $\mathbb{E}_{\mathbf{u}\sim P_t}\big[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))\big] \leq \exp\big(-\eta\widetilde{\ell}_t(\mathbf{w}_t)\big) = 1$ for each iteration. This can be achieved by the following arguments

$$\begin{aligned} \mathbb{E}_{\mathbf{u}\sim P_t}\Big[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))\Big] &= \sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\mathbb{E}_{\mathbf{u}\sim P_{t,i}}[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))] \\ &= \sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\mathbb{E}_{\mathbf{u}\sim P_{t,i}}\Big[\exp\Big(\eta\mathbf{g}_t^\top(\mathbf{w}_t - \mathbf{u}) - \eta^2\|\mathbf{u} - \mathbf{w}_t\|^2_{\mathbf{g}_t\mathbf{g}_t^\top}\Big)\Big] \\ &\leq \sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\exp\Big(\eta\mathbf{g}_t^\top(\mathbf{w}_t - \mathbf{w}_{t,i}) - \eta^2\|\mathbf{w}_t - \mathbf{w}_{t,i}\|^2_{\mathbf{g}_t\mathbf{g}_t^\top}\Big) \\ &\leq \sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\big(1 + \eta\mathbf{g}_t^\top(\mathbf{w}_t - \mathbf{w}_{t,i})\big) = 1, \end{aligned}$$

where the first inequality is due to [van Erven and Koolen, 2016, Lemma 10] under the condition $\eta \leq 1/(10G)$. The second inequality holds because $e^{z-z^2} \leq 1 + z$ for any $z \geq -\frac{2}{3}$. Then, we can have

$$\text{term (a)} \leq 0. \tag{27}$$

**Bounding term (c).** A direct calculation according to the definition of $\widetilde{\ell}_t$ shows that

$$\begin{aligned} \texttt{term (c)} &= \sum_{t=1}^{T}\mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] - \sum_{t=1}^{T}\widetilde{\ell}_t(\mathbf{u}_t) \\ &= \eta\sum_{t=1}^{T}\mathbb{E}_{\mathbf{u}\sim Q_t}[(\mathbf{g}_t^\top(\mathbf{u} - \mathbf{u}_t))^2] \\ &= \eta\sigma^2\sum_{t=1}^{T}\|\mathbf{g}_t\|_2^2 \leq \frac{dGT\sigma^2}{2}, \end{aligned} \tag{28}$$

where the last inequality is due to $\|\mathbf{g}_t\|_2 \leq \sqrt{d}\|\mathbf{g}_t\|_\infty \leq \sqrt{d}G$ and $\eta \leq \frac{1}{2G}$.

**Bounding term (b).** As for term (b), different from the previous work [Zhang et al., 2025], we decompose the mix loss by exploiting the two-layer structure. Denote by

$$\widetilde{m}_t(P_{t,i}) = -\frac{1}{\eta}\ln\Big(\mathbb{E}_{P_{t,i}}[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))]\Big)$$

the mix loss for the individual distribution $P_{t,i}$. We can rewrite the mix loss for the aggregated distribution $P_t$ as

$$\begin{aligned}\widetilde{m}_t(P_t) &= -\frac{1}{\eta}\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\mathbb{E}_{\mathbf{u}\sim P_{t,i}}[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))]\right)\\ &= -\frac{1}{\eta}\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-\eta\widetilde{m}_t(P_{t,i}))\right)\\ &= \sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\cdot\widetilde{m}_t(P_{t,i}) + \frac{1}{\eta}\left(\mathrm{KL}\left(\mathbf{q}_t\,\|\,\mathbf{p}_t\right) - \mathrm{KL}\left(\mathbf{q}_t\,\|\,\widetilde{\mathbf{p}}_{t+1}\right)\right), \end{aligned} \tag{29}$$

where $\mathbf{p}_t \in \Delta_{|\mathcal{H}_t|}$ denotes the probability vector over the base-learner pool with the $i$-th entry $p_{t,i}$ and $\widetilde{p}_{t+1,i} \propto p_{t,i}\cdot\exp(-\eta\widetilde{m}_t(P_{t,i})) = p_{t,i}\cdot\mathbb{E}_{P_{t,i}}[\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))]$ is the same as the one defined in Algorithm 2. The last line holds for any $\mathbf{q}_t \in \Delta_{|\mathcal{H}_t|}$ that assigns weight to each elements in $\mathcal{H}_t$ and $q_{t,i}$ is the $i$-th entry of $\mathbf{q}_t$. Furthermore, we can also rewrite the mix loss for each base learner $\mathcal{B}_i$ as

$$\begin{aligned}\widetilde{m}_t(P_{t,i}) &= \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\left(\mathrm{KL}\left(Q_t\,\|\,P_{t,i}\right) - \mathrm{KL}\left(Q_t\,\|\,P'_{t+1,i}\right)\right)\\ &\le \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\left(\mathrm{KL}\left(Q_t\,\|\,P_{t,i}\right) - \mathrm{KL}\left(Q_t\,\|\,P_{t+1,i}\right)\right), \end{aligned} \tag{30}$$

where $P'_{t+1,i} \propto P_{t,i}\exp(-\eta\widetilde{\ell}_t(\mathbf{u}))$ is the same as (5) and the last line is to the Pythagorean theorem for KL divergence since $P_{t+1,i} = \arg\min_{P'\in\mathscr{W}} \mathrm{KL}\left(P'\,\|\,P'_{t+1,i}\right)$ and $\mathscr{W}$ is a convex set. Then, plugging (30) back into (29), we arrive

$$\begin{aligned}\widetilde{m}_t(P_t) \le \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] &+ \underbrace{\frac{1}{\eta}\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\mathrm{KL}\left(Q_t\,\|\,P_{t,i}\right) + \mathrm{KL}\left(\mathbf{q}_t\,\|\,\mathbf{p}_t\right)\right)}_{\texttt{term (b-1)}}\\ &\underbrace{-\frac{1}{\eta}\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\mathrm{KL}\left(Q_t\,\|\,P_{t+1,i}\right) + \mathrm{KL}\left(\mathbf{q}_t\,\|\,\widetilde{\mathbf{p}}_{t+1}\right)\right)}_{\texttt{term (b-2)}} \end{aligned} \tag{31}$$

holds for any $\mathbf{q}_t \in \Delta_{|\mathcal{H}_t|}$ and $Q_t$. Here, we specify $\mathbf{q}_t$ as the minimizer of the optimization problem

$$\mathbf{q}_t = \arg\min_{\mathbf{q}\in\Delta_{|\mathcal{H}_t|}} \sum_{\mathcal{B}_i\in\mathcal{H}_t} q_i\mathrm{KL}\left(Q_t\,\|\,P_{t,i}\right) + \mathrm{KL}\left(\mathbf{q}\,\|\,\mathbf{p}_t\right),$$

whose optimal value has the close form formulation as

$$V_t(Q_t) = -\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-\mathrm{KL}\left(Q_t\,\|\,P_{t,i}\right))\right) \ge 0.$$

The value $V_t(Q_t)$ is always greater than 0 since the objective function of the above optimization problem is non-negative. Then, we have

$$\texttt{term (b-1)} \le \frac{1}{\eta}V_t(Q_t).$$

As for term (b-2), we can similarly define

$$\widetilde{\mathbf{q}}_{t+1} = \arg\min_{\mathbf{q}\in\Delta_{|\mathcal{H}_t|}} \sum_{\mathcal{B}_i\in\mathcal{H}_t} q_i\mathrm{KL}\left(Q_t\,\|\,P_{t+1,i}\right) + \mathrm{KL}\left(\mathbf{q}\,\|\,\widetilde{\mathbf{p}}_{t+1}\right).$$

We also have $\widetilde{V}_{t+1}(Q_t) = -\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t}\widetilde{p}_{t+1,i}\cdot\exp(-\mathrm{KL}\left(Q_t\,\|\,P_{t+1,i}\right))\right)$ as the optimal value of the above optimization problem. Clearly, we have

$$\texttt{term (b-2)} = -\frac{1}{\eta}\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} q_{t,i}\mathrm{KL}\left(Q_t\,\|\,P_{t+1,i}\right) + \mathrm{KL}\left(\mathbf{q}_t\,\|\,\widetilde{\mathbf{p}}_{t+1}\right)\right) \le -\frac{1}{\eta}\widetilde{V}_{t+1}(Q_t),$$

since $\mathbf{q}_{t,i}$ is not the minimizer of the objective function in term (b-2). Plugging the upper bound of term (b-1) and term (b-2) into (31), we have

$$\widetilde{m}_t(P_t) \leq \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\left(V_t(Q_t) - \widetilde{V}_{t+1}(Q_t)\right). \tag{32}$$

Then, we related $\widetilde{V}_{t+1}(Q_t)$ to $V_{t+1}(Q_t)$ by

$$\begin{aligned}
V_{t+1}(Q_t) =& -\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_{t+1}} p_{t+1,i}\cdot\exp(-\mathrm{KL}\left(Q_t \,\|\, P_{t+1,i}\right))\right)\\
=& -\ln\left((1-\mu_{t+1})\sum_{\mathcal{B}_i\in\mathcal{H}_t}\widetilde{p}_{t+1,i}\cdot\exp(-\mathrm{KL}\left(Q_t \,\|\, P_{t+1,i}\right)) + \mu_{t+1}\exp(-\mathrm{KL}\left(Q_t \,\|\, N_0\right))\right)\\
\leq& -\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t}\widetilde{p}_{t+1,i}\cdot\exp(-\mathrm{KL}\left(Q_t \,\|\, P_{t+1,i}\right))\right) + \ln\left(\frac{1}{1-\mu_{t+1}}\right)\\
=& \widetilde{V}_{t+1}(Q_t) + \frac{\mu_{t+1}}{1-\mu_{t+1}} \leq \widetilde{V}_{t+1}(Q_t) + \frac{1}{t},
\end{aligned} \tag{33}$$

where the second line is due to the fixed-share update (7) with $N_0 = \mathcal{N}(\mathbf{u}_0, I_d)$ and the last inequality is due to the parameter setting $\mu_{t+1} = 1/(t+1)$. Plugging (33) back into (32) and taking a summation over $T$ rounds, we obtain

$$\begin{aligned}
\sum_{t=1}^{T}\widetilde{m}_t(P_t) =& \sum_{t=1}^{T}\mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\sum_{t=1}^{T}\left(V_t(Q_t) - V_{t+1}(Q_t)\right) + \sum_{t=1}^{T}\frac{1}{\eta t}\\
\leq& \sum_{t=1}^{T}\mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] + \frac{1}{\eta}\sum_{t=2}^{T}\left(V_t(Q_t) - V_t(Q_{t-1})\right) + \frac{1}{\eta}V_1(Q_1) + \frac{2}{\eta}\log T.
\end{aligned} \tag{34}$$

where the first inequality holds because $V_t(Q_t)$ is always non-negative and $\sum_{t=1}^{T}\frac{1}{t+1} \leq \log T$.

It remains to handle the variation term $V_t(Q_t) - V_t(Q_{t-1})$. Denote by

$$h_{t,i}(\mathbf{u}) = \frac{1}{2}\left(\log|H_{t,i}^{-1}| + \sigma^2\mathrm{Tr}(H_{t,i}) + \|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}^2\right), \tag{35}$$

where $\mathrm{Tr}(A)$ indicates the trace of a matrix $A$. Then, the variation term can be expressed as

$$\begin{aligned}
V_t(Q_t) - V_t(Q_{t-1}) =& \ln\left(\frac{\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-\mathrm{KL}\left(Q_{t-1} \,\|\, P_{t,i}\right))}{\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-\mathrm{KL}\left(Q_t \,\|\, P_{t,i}\right))}\right)\\
=& \ln\left(\frac{\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-h_{t,i}(\mathbf{u}_{t-1}))}{\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-h_{t,i}(\mathbf{u}_t))}\right)\\
=& J_t(\mathbf{u}_t) - J_t(\mathbf{u}_{t-1}) \leq \sup_{\mathbf{u}\in\mathcal{W}}\|\nabla J_t(\mathbf{u})\|_2\cdot\|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2
\end{aligned} \tag{36}$$

where the second line is due to the definition of KL divergence for Gaussian distributions and we define $J_t(\mathbf{u}) = -\ln\left(\sum_{\mathcal{B}_i\in\mathcal{H}_t} p_{t,i}\cdot\exp(-h_{t,i}(\mathbf{u}))\right)$ in the last line. The next step is to control the gradient of the function $J_t(\mathbf{u})$, which can be calculated as

$$\nabla J_t(\mathbf{u}) = \sum_{\mathcal{B}_i\in\mathcal{H}_t}\beta_{t,i}(\mathbf{u})H_{t,i}(\mathbf{u}-\mathbf{w}_{t,i}) \quad \forall\mathbf{u}\in\mathcal{W},$$

where $\beta_{t,i}(\mathbf{u}) \in \Delta_{|\mathcal{H}_t|}$ and $\beta_{t,i}(\mathbf{u}) \propto p_{t,i}\exp(-h_{t,i}(\mathbf{u}))$. We then bound the norm of $\nabla J_t(\mathbf{u})$ by

$$\begin{aligned}
\|\nabla J_t(\mathbf{u})\|_2 \leq& \sum_{\mathcal{B}_i\in\mathcal{H}_t}\beta_{t,i}(\mathbf{u})\cdot\|H_{t,i}(\mathbf{u}-\mathbf{w}_{t,i})\|_2\\
\leq& \sum_{\mathcal{B}_i\in\mathcal{H}_t}\beta_{t,i}(\mathbf{u})\cdot\sqrt{\mathrm{Tr}(H_{t,i})}\cdot\|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}\\
\leq& \sqrt{\sum_{\mathcal{B}_i\in\mathcal{H}_t}\beta_{t,i}(\mathbf{u})\|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}^2}\sqrt{\sum_{\mathcal{B}_i\in\mathcal{H}_t}\beta_{t,i}\mathrm{Tr}(H_{t,i})}
\end{aligned} \tag{37}$$

where the second inequality is by Cauchy–Schwarz inequality and $\beta_{t,i}(\mathbf{u}) \in \Delta_{|\mathcal{H}_t|}$. The third line holds because $\|H_{t,i}(\mathbf{u}-\mathbf{w}_{t,i})\|_2^2 \le \|H_{t,i}\|_2 \|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}^2 \le \mathrm{Tr}(H_{t,i})\|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}^2$ for a symmetric positive definite matrix $H_{t,i}$.

Then, we proceed to relate the above two terms back to $h_{t,i}(\mathbf{u})$. As shown in (26), $H_{t,i} = I_d + 2\eta^2 \sum_{t=i}^{t} g_t g_t^\top \preceq \left(1+\frac{td}{2}\right) I_d \preceq dT I_d$ for any $\mathcal{B}_i \in \mathcal{H}_t$ and $t \in [T]$, which implies $\lambda_{\min}(H_{t,i}^{-1}) \ge 1/(dT)$. Consequently, one has $\log|H_{t,i}^{-1}| \ge -d\log(dT)$. Plugging the lower bound into (35) yields

$$\mathrm{Tr}(H_{t,i}) \le \frac{2h_{t,i}(\mathbf{u}) + d\log(dT)}{\sigma^2} \quad \text{and} \quad \|\mathbf{u}-\mathbf{w}_{t,i}\|_{H_{t,i}}^2 \le 2h_{t,i}(\mathbf{u}) + d\log(dT). \tag{38}$$

Then, plugging (38) into (37), we can further upper bound the gradient norm by

$$\begin{aligned}
\|\nabla J_t(\mathbf{u})\|_2 &\le \frac{1}{\sigma}\left(d\log(dT) + 2\sum_{\mathcal{B}_i\in\mathcal{H}_t} \beta_{t,i}(\mathbf{u}) h_{t,i}(\mathbf{u})\right) \\
&\le \frac{1}{\sigma}\left(d\log(dT) + 2\sum_{\mathcal{B}_i\in\mathcal{H}_t} \beta_{t,i}(\mathbf{u}) h_{t,i}(\mathbf{u}) + 2\mathrm{KL}\left(\beta_t(\mathbf{u}) \,\|\, \mathbf{p}_t\right)\right) \\
&= \frac{1}{\sigma}\left(d\log(dT) - 2\ln\left(\sum_{\mathcal{B}_t\in\mathcal{H}_t} p_{t,i}\exp(-h_{t,i}(\mathbf{u}))\right)\right) \\
&\le \frac{1}{\sigma}\left(d\log(dT) - 2\ln\left(\mu_t \cdot e^{-h_{t,t}(\mathbf{u})}\right)\right) \\
&= \frac{1}{\sigma}(d\log(dT) + 2\ln t + \sigma^2 d + \|\mathbf{u}-\mathbf{u}_0\|_2^2) \\
&\le \frac{(d+2)\log(dT) + d + 4}{\sigma}
\end{aligned} \tag{39}$$

where the third line is by the definition $\beta_{t,i}(\mathbf{u}) \propto p_{t,i}\exp(-h_{t,i}(\mathbf{u}))$. The fourth line is due to the fixed share update (7) such that there always exists a base algorithm with weight $\mu_t = 1/t$ and distribution $P_{t,t} = N_0 = \mathcal{N}(\mathbf{u}, I_d)$. The last line is by the condition $\sigma \in (0,1]$ and $\|\mathbf{u}-\mathbf{u}_0\|_2 \le \|\mathbf{u}-\mathbf{u}_0\|_1 \le 2$ for any $\mathbf{u} \in \mathcal{W} = \Delta_d$. Finally, we can upper bound term (B) by

$$\begin{aligned}
\texttt{term (b)} &= \sum_{t=1}^{T} \widetilde{m}_t(P_t) - \sum_{t=1}^{T} \mathbb{E}_{\mathbf{u}\sim Q_t}[\widetilde{\ell}_t(\mathbf{u})] \\
&\le \frac{1}{\eta}\frac{((d+2)\log(dT)+d+4)\sum_{t=2}^{T}\|\mathbf{u}_t-\mathbf{u}_{t-1}\|_2}{\sigma} + \frac{1}{\eta}V_1(Q_1) + \frac{2}{\eta}\log(dT) \\
&= \frac{1}{\eta}\frac{((d+2)\log(dT)+d+4)P_T}{\sigma} + \frac{1}{\eta}\mathrm{KL}(Q_1\|P_1) + \frac{2}{\eta}\log(dT) \\
&\le \frac{((d+2)\log(dT)+d+4)P_T}{\eta\sigma} + \frac{1}{\eta}\left(3 + d\log\left(\frac{1}{\sigma}\right) + \frac{d\sigma^2}{2} + \log(dT)\right),
\end{aligned} \tag{40}$$

where the second line comes from a combination of (34), (36) and (39). The second line is by the definition of $V_1(Q_1)$ and the last line is due to the close formulation of the KL divergence between two Gaussian distributions.

**Combining All.** Combining the upper bounds (27), (28) and (40) on term (a), term (b) and term (c), we arrive

$$\begin{aligned}
&\sum_{t=1}^{T}\ell_t(\mathbf{w}_t) - \sum_{t=1}^{T}\ell_t(\mathbf{u}_t) \\
&\le \frac{((d+2)\log(dT)+d+4)P_T}{\eta\sigma} + \frac{1}{\eta}\left(3 + d\log\left(\frac{1}{\sigma}\right) + \frac{d\sigma^2}{2} + \log T\right) + \frac{dGT\sigma^2}{2}.
\end{aligned}$$

We can further bound the above regret depending on different value of $P_T$.

- **Case 1:** $P_T \le 1/\sqrt{T}$, we choose $\sigma = \sqrt{\frac{1}{T}}$ to obtain $\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}(Gd\log T + G)$.
- **Case 2:**, we can choose $\sigma = \min\{P_T^{\frac{1}{3}} T^{-\frac{1}{3}}, 1\}$ and obtain

$$\text{D-Reg}(\{\mathbf{u}_t\}_{t=1}^T) \le \mathcal{O}\left(dG\log T(1 + T^{\frac{1}{3}} P_T^{\frac{2}{3}})\right).$$

We complete the proof by combining the two cases.

□

# E Technical Lemmas

**Lemma 5.** *Let $\psi(\mathbf{u}) = \frac{\mathrm{d}}{\mathrm{d}u}\ln\Gamma(u)$ be the digamma function. Then, $g(u) = \gamma u\psi(1+\gamma u) - \ln\Gamma(1+\gamma u)$ is a convex function for $u > 0$ and $\gamma > 0$.*

*Proof of Lemma 5.* Let $\psi(u) = \frac{\mathrm{d}}{\mathrm{d}u}\ln\Gamma(u)$ and define $g(u) = \gamma u\psi(1+\gamma u) - \ln\Gamma(1+\gamma u)$ for $\gamma > 0$. Set $x = 1+\gamma u$ and define $h(x) = (x-1)\psi(x) - \ln\Gamma(x)$ for $x > 1$. Since $g(u) = h(1+\gamma u)$, it suffices to show that $h$ is convex on $(1,\infty)$.

A direct computation yields $h'(x) = (x-1)\psi'(x)$ and $h''(x) = \psi'(x) + (x-1)\psi''(x)$. Using the integral representations of the polygamma functions, we have

$$\psi'(x) = \int_0^\infty \frac{te^{-xt}}{1-e^{-t}}\,\mathrm{d}t, \qquad \psi''(x) = -\int_0^\infty \frac{t^2e^{-xt}}{1-e^{-t}}\,\mathrm{d}t,$$

for $x > 1$. Then, by the integration by parts arguments, we obtain for $x > 1$,

$$\begin{aligned}
h''(x) &= \int_0^\infty \frac{t}{e^t-1}\,(1-(x-1)t)\,e^{-(x-1)t}\,\mathrm{d}t \\
&= \int_0^\infty \frac{t}{e^t-1}\,\frac{\mathrm{d}}{\mathrm{d}t}\Big(te^{-(x-1)t}\Big)\,\mathrm{d}t \\
&= \left[\frac{t^2e^{-(x-1)t}}{e^t-1}\right]_0^\infty - \int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{e^t-1}\right)te^{-(x-1)t}\,\mathrm{d}t \\
&= -\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{e^t-1}\right)te^{-(x-1)t}\,\mathrm{d}t.
\end{aligned}$$

The last equality holds because the function $\frac{t^2e^{-(x-1)t}}{e^t-1} \to 0$ when $t \to 0$ and $t \to \infty$.

Since $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{e^t-1}\right) < 0$ for $t > 0$, the integrand is nonnegative and not identically zero. Hence $h''(x) > 0$ for all $x > 1$, so $h$ is strictly convex. Therefore $g(u) = h(1+\gamma u)$ is strictly convex for $u > 0$ and $\gamma > 0$. □